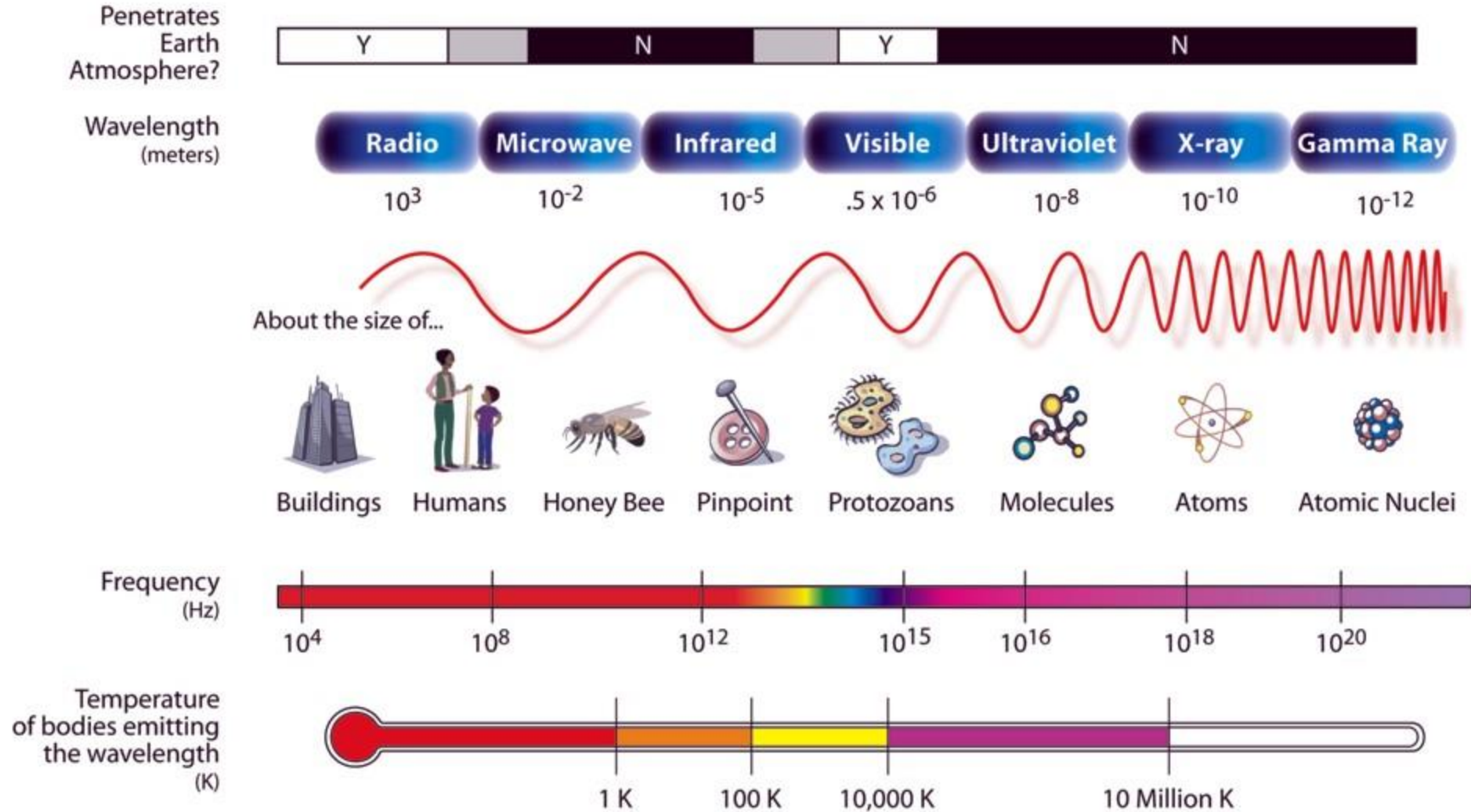


THE ELECTROMAGNETIC SPECTRUM



Discovery of X-rays and diffraction



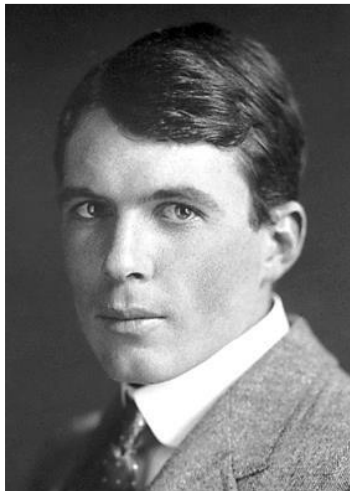
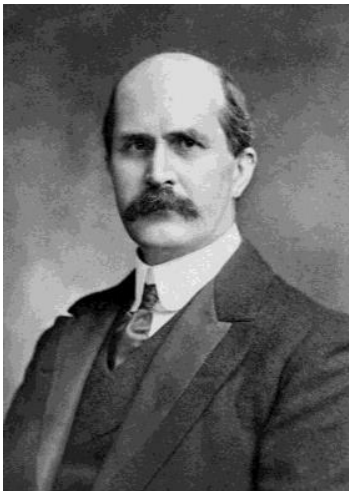
Discovery in 1895

Wilhem Conrad Röntgen
(Nobel prize 1901)



X-ray diffraction on a single crystal 1912

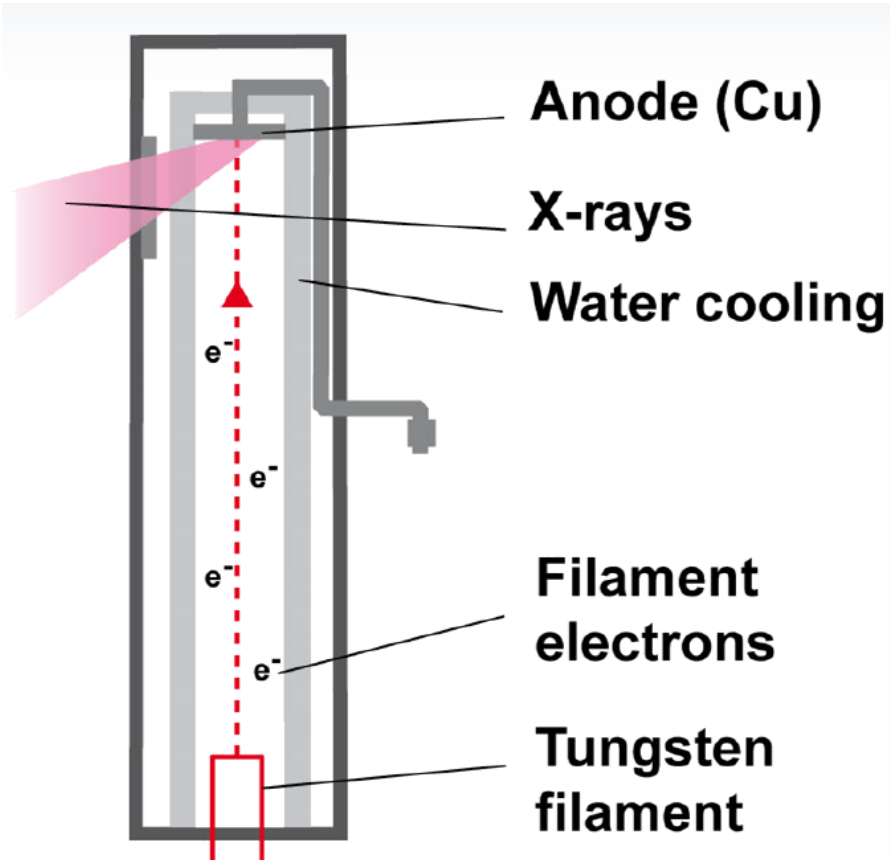
Friedrich, Knipping and von Laue
(Nobel prize 1914)



Analysis of crystal structures using X-rays

William Henry Bragg & William Laurence Bragg
(Nobel prize 1915)

X-ray tube

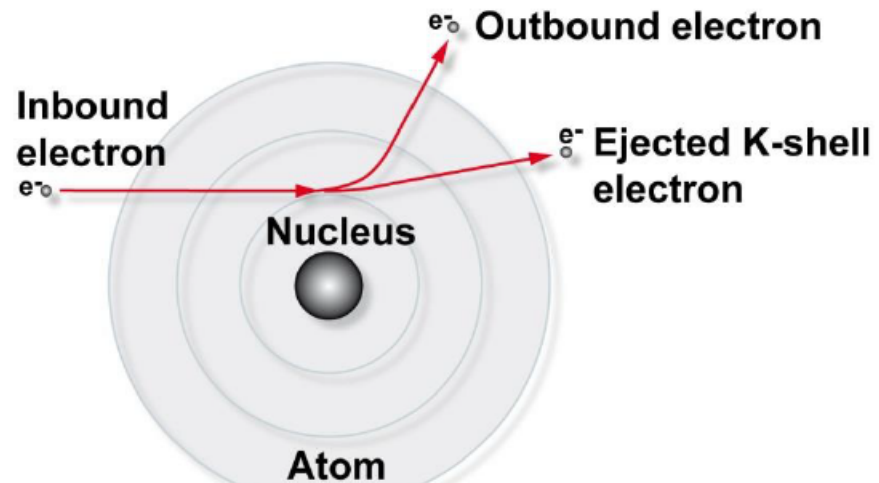


- Current (mA) passes through the W filament
 - Emission of electrons
- High voltage (kV) is applied across filament and anode
 - Electrons are accelerated towards the metal anode (target and filament in vacuum)
- Fast electrons hit metal anode and are decelerated
 - Heat & emission of X-rays (target must be cooled)

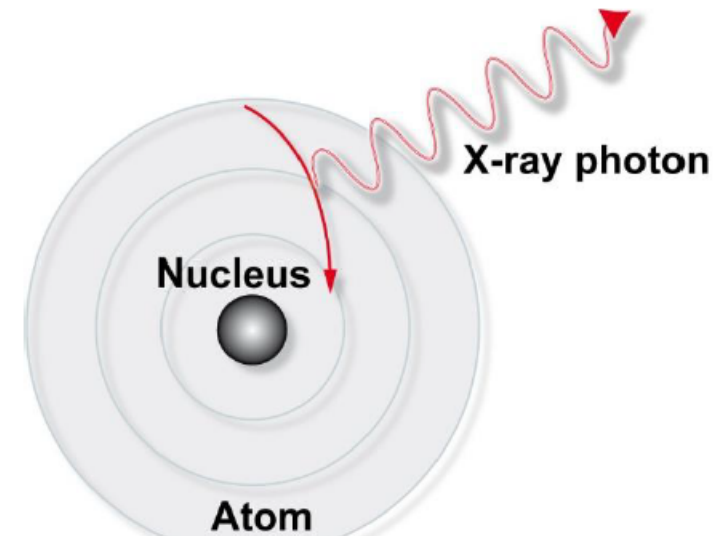
X-rays production (emission)

- Electromagnetic radiation (X-rays) is produced when a charged particle (electron) changes speed
- Electrons from the filament collide with anode material and are decelerated

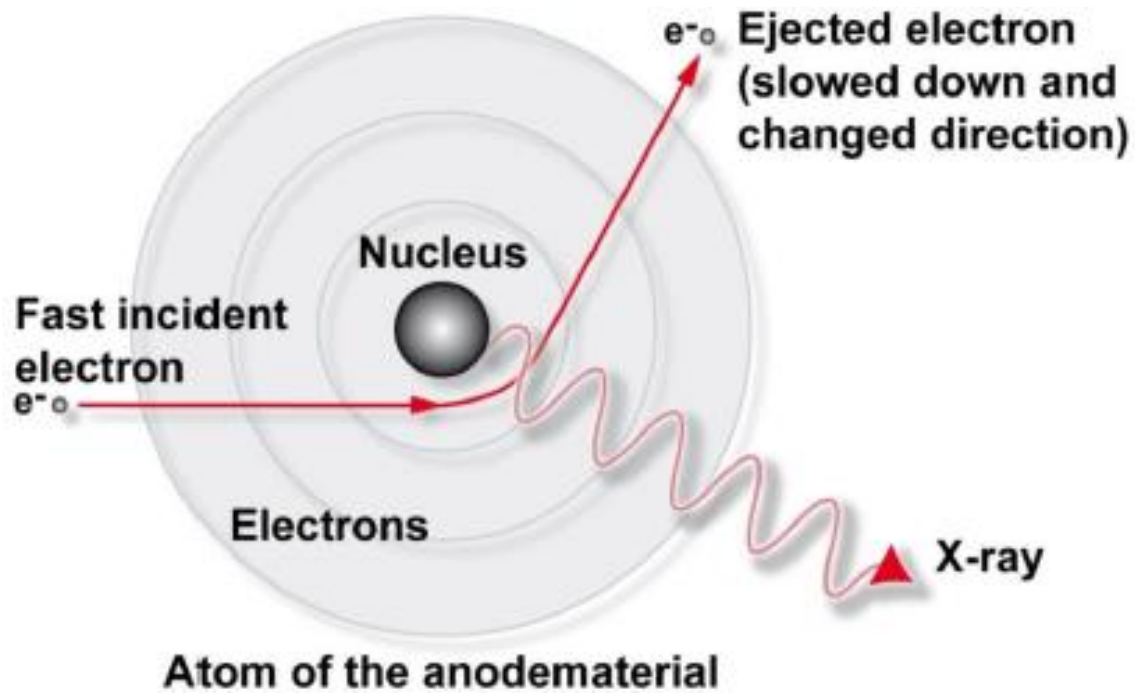
Producing heat and **continuous 'Bremsstrahlung'**



Producing **characteristic radiation** via the photoelectric effect



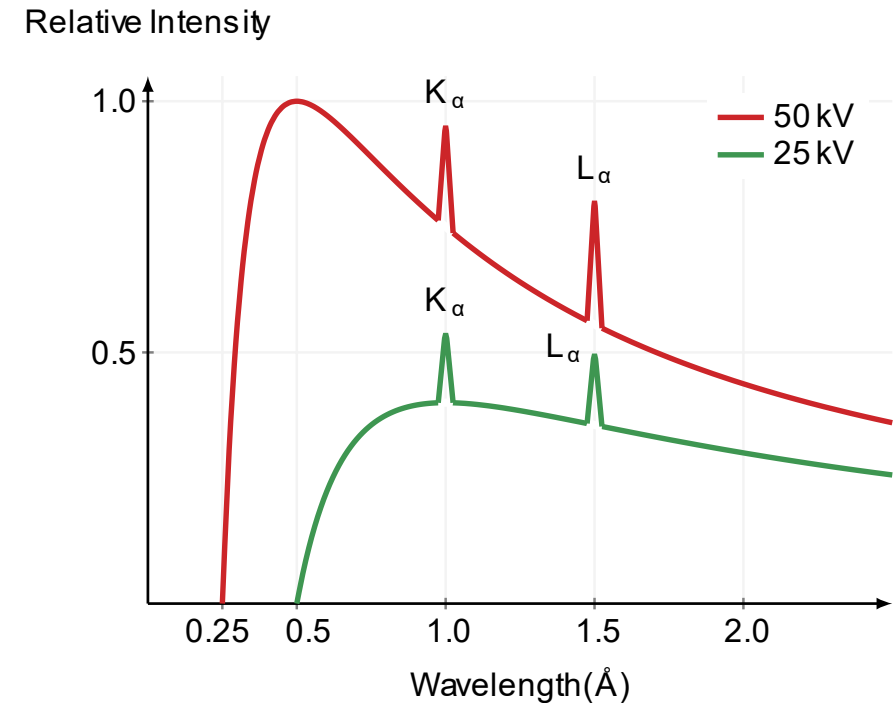
Continuous radiation



When an electron passes near the nucleus it is slowed and its path is deflected. Energy lost is emitted as a Bremsstrahlung X-ray photon.

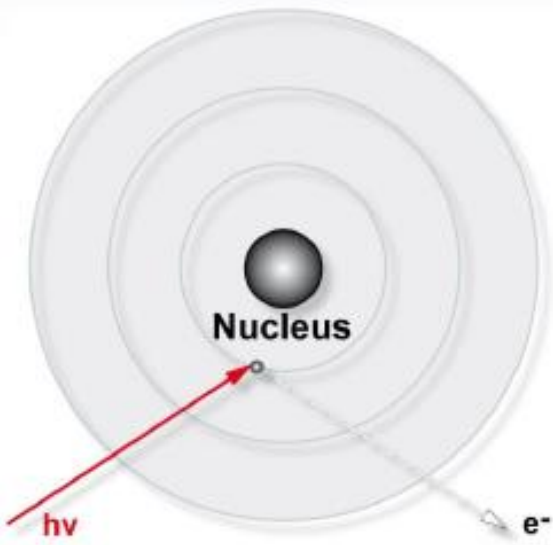
$$\lambda_{min} = \frac{hc}{eV} \approx \frac{12.4}{V(kV)}$$

shortest wavelength of continuous radiation

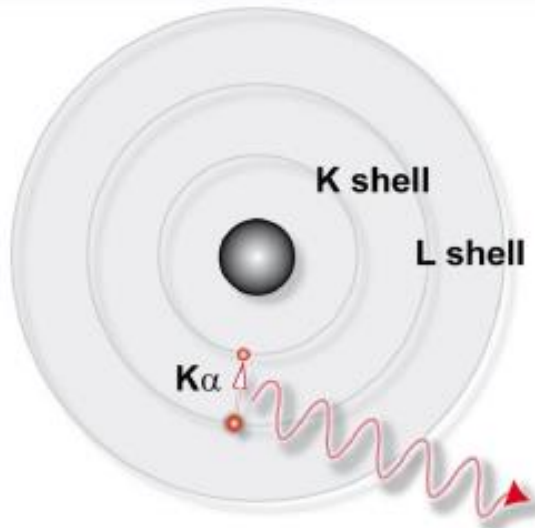


Bremsstrahlung, white radiation

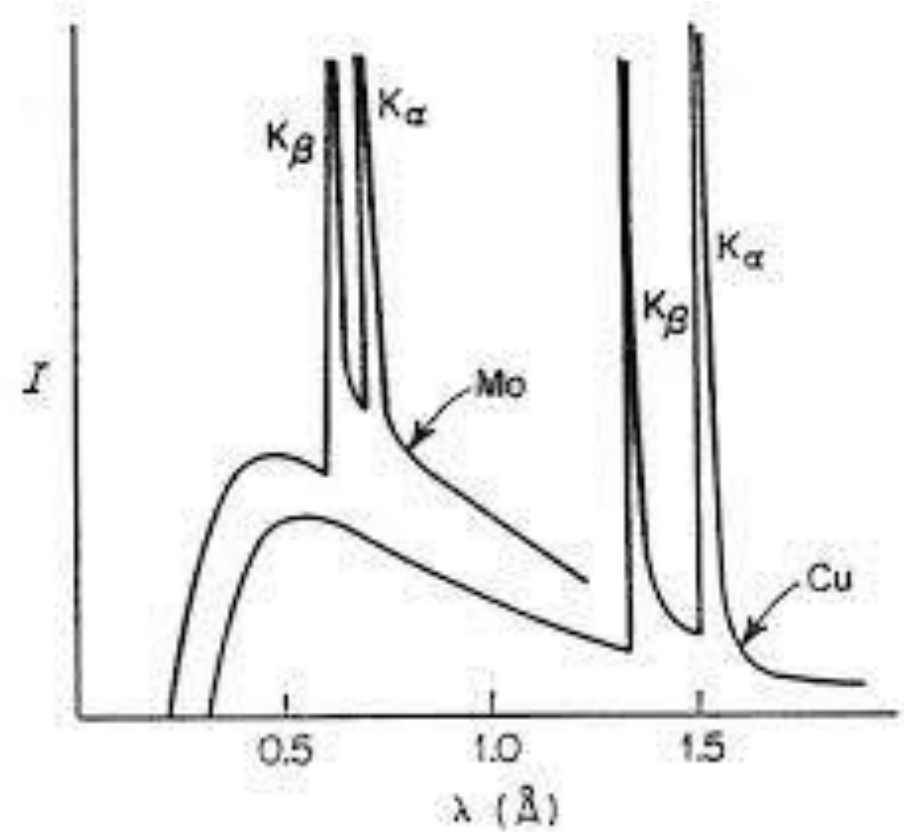
Characteristic X-rays



- High-voltage electrons can ionize inner shell electrons, creating a vacancy



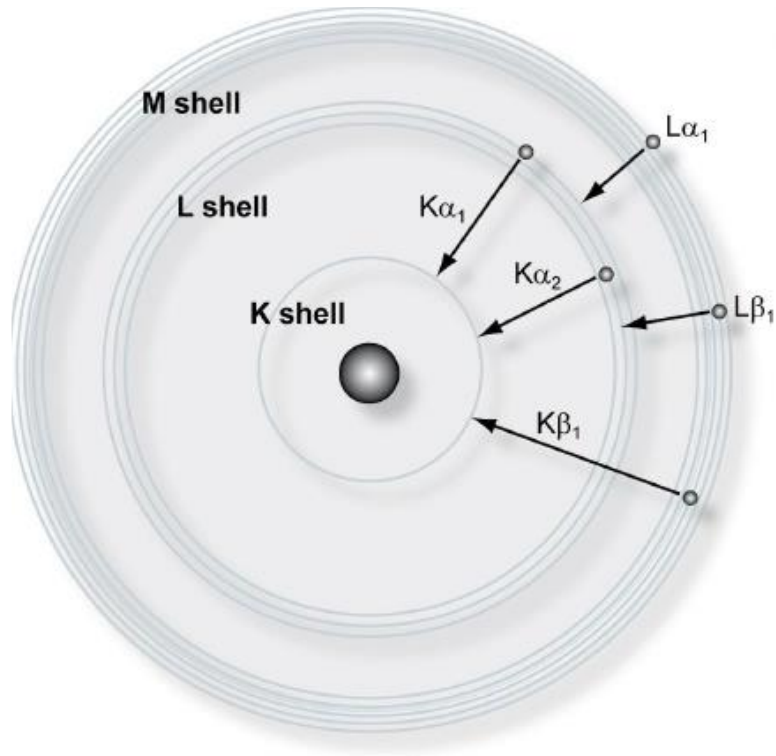
- When electrons from outer shells fill these vacancies, they release radiation



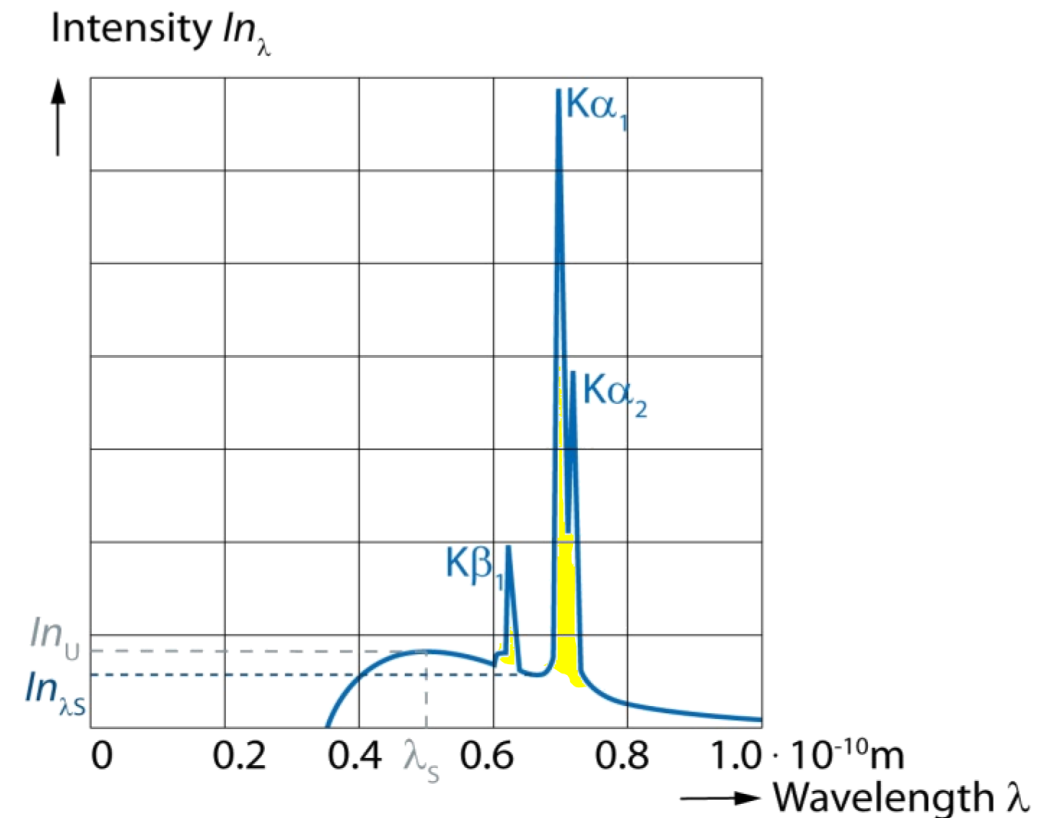
Characteristic X-rays

The anode material determines the wavelengths of the characteristic radiation.

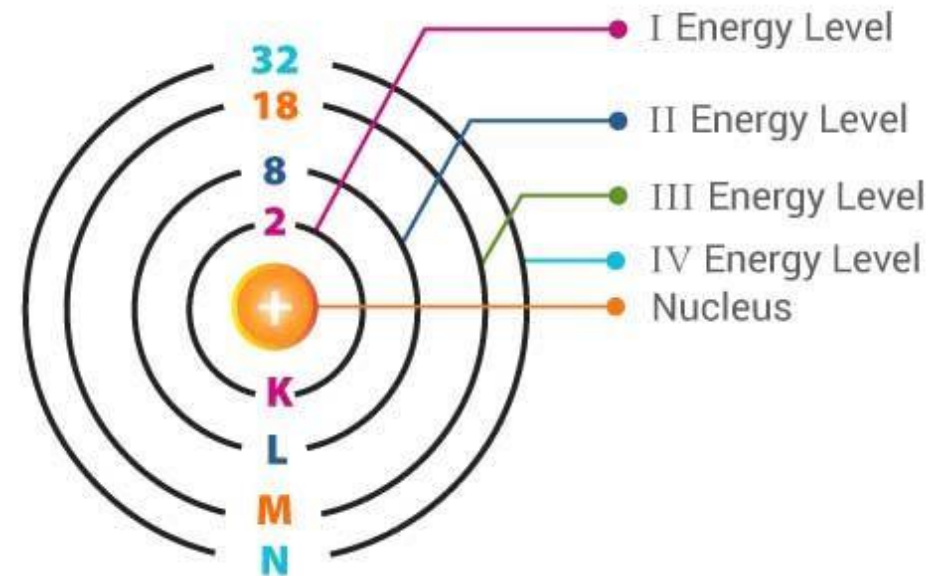
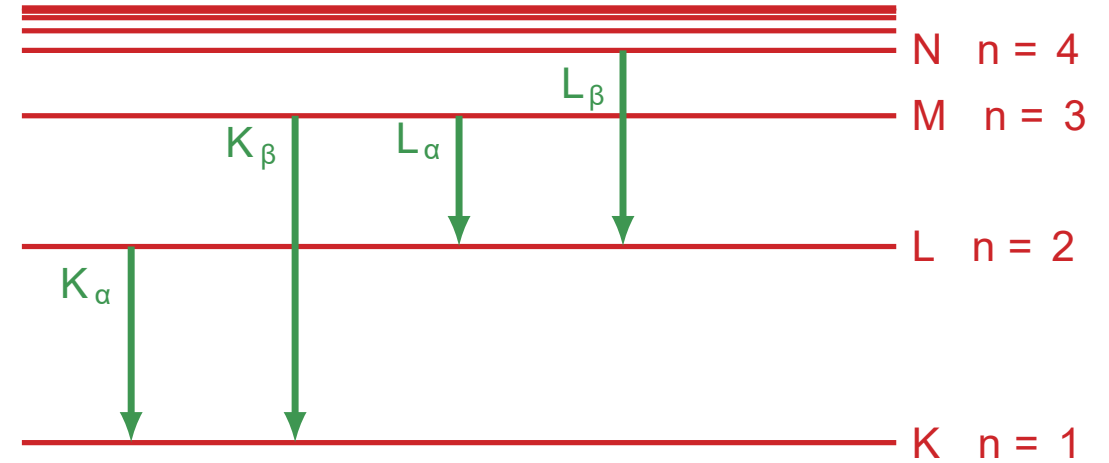
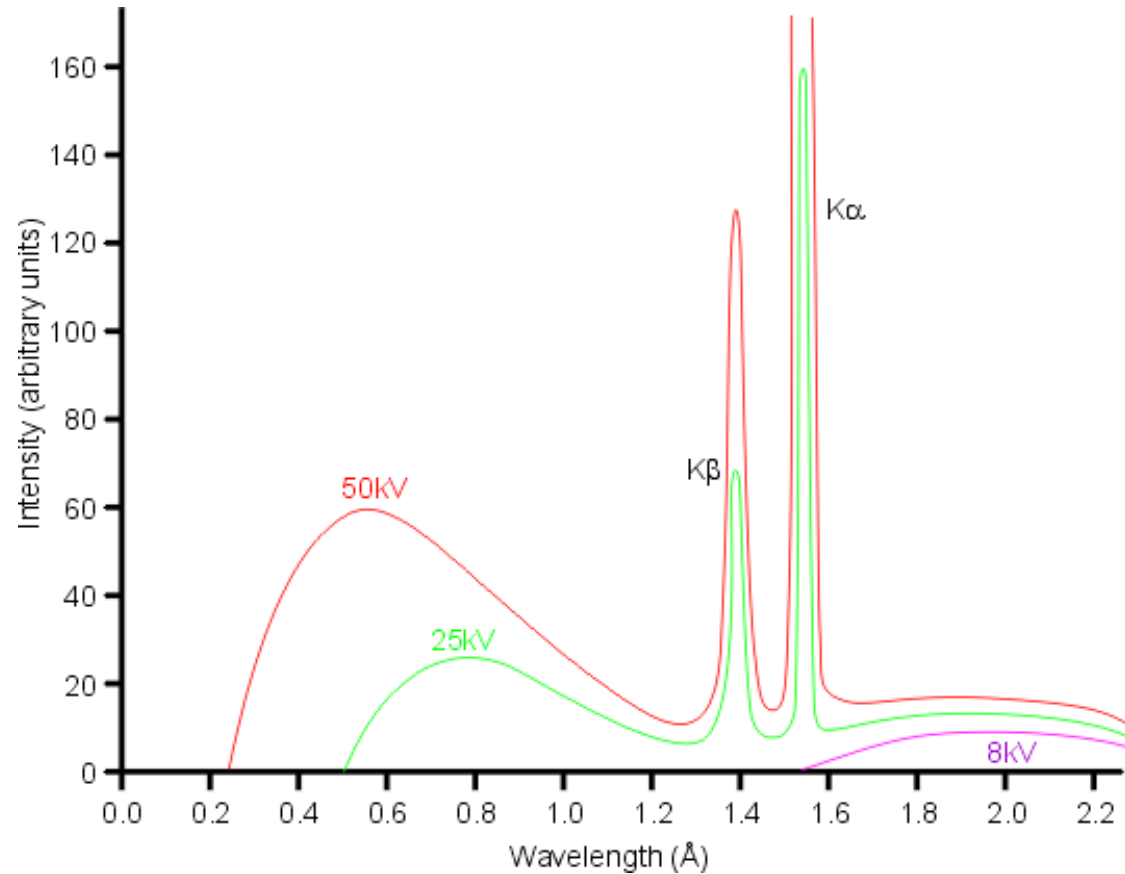
$$K_{\alpha 1} : K_{\alpha 2} : K_{\beta}$$
$$100 : 50 : 20$$



When a high energy electron collides with an inner shell electron both are ejected from the atom leaving a 'hole' in the inner layer. This is filled by an outer shell electron with a loss of energy emitted as an X-ray photon.



Characteristic X-rays

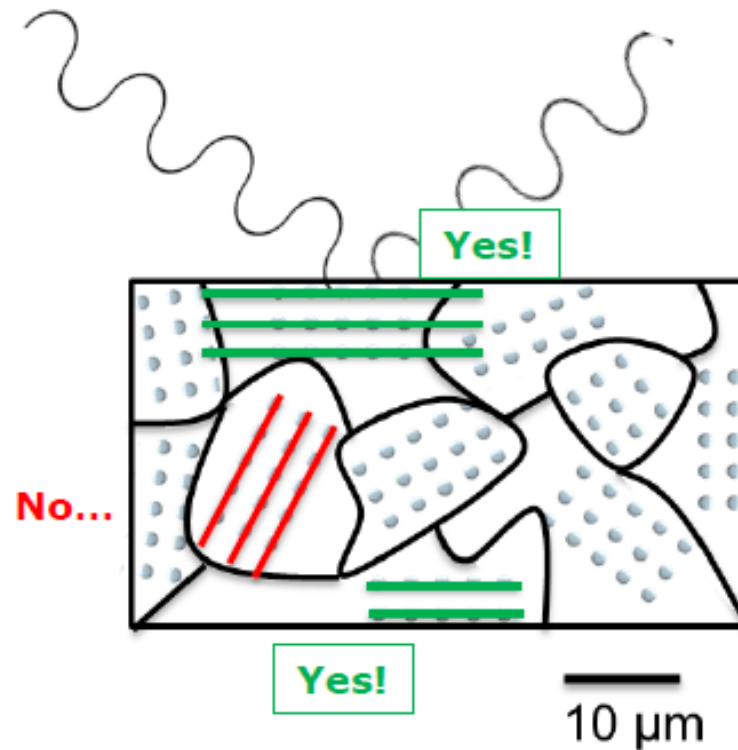


Typical wavelength

Anode	Activation [kV]	Wavelength	λ [Å]	Filter	[mm]
Mo	20.0	$k\alpha_1$	0.7093187(4)	Zr	0.081
		$k\alpha_2$	0.713609(6)		
		$k\beta_1$	0.632305(9)		
Cu	8.981	$k\alpha_1$	1.540598(2)	Ni	0.015
		$k\alpha_2$	1.544426(2)		
		$k\beta_1$	1.39225(1)		
Co	7.709	$k\alpha_1$	1.78901(1)	Fe	0.012
		$k\alpha_2$	1.79290(1)		
		$k\beta_1$	1.62083(2)		
Fe	7.111	$k\alpha_1$	1.93609(1)	Mn	0.011
		$k\alpha_2$	1.94003(1)		
		$k\beta_1$	1.75665(2)		
Cr	5.989	$k\alpha_1$	2.28976(2)	V	0.011
		$k\alpha_2$	2.293663(6)		
		$k\beta_1$	2.08492(2)		

The interaction of X-rays and matter

Diffraction from planes of atoms parallel to surface



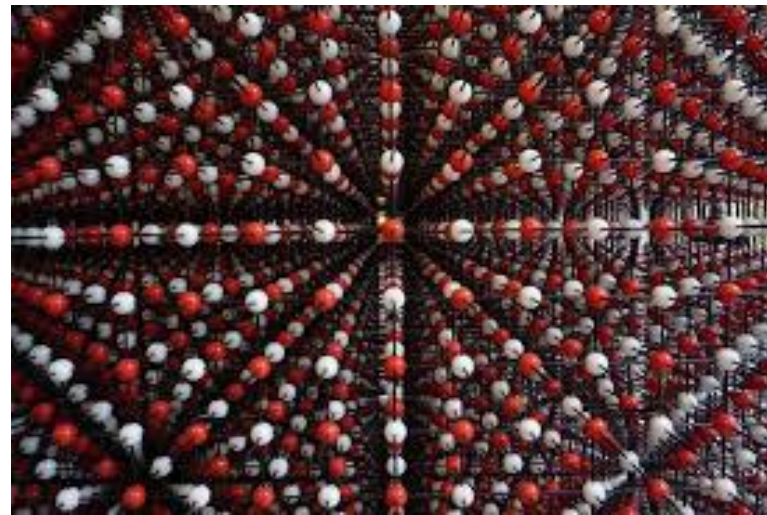
What are lattice planes?

What is a crystal?

- A crystal is an ordered arrangement of atoms
- To describe a crystal, we would need to know the relative positions of all atoms



NaCl crystal

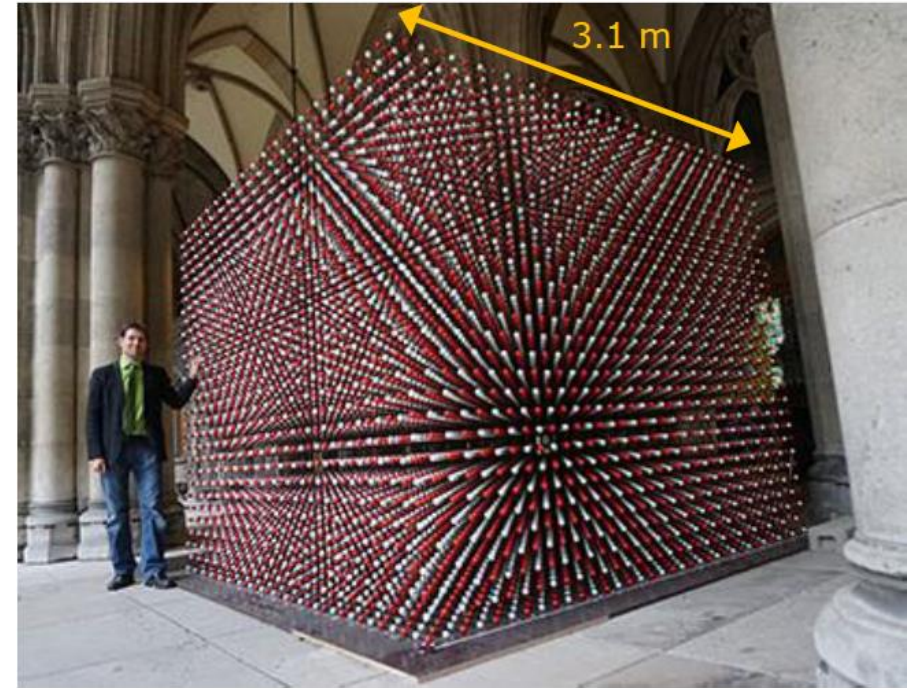


Model structure of NaCl

- The problem: a tiny crystal contains a huge number of atoms!

What is a crystal structure?

- A grain of salt (edge = 0.0000096 mm) contains $\sim 40,000$ atoms!
- Magnified by ca. 1 billion, this grain would look like this →
- Clearly, it would take very long to write down the positions of all atoms even for such a small crystal.

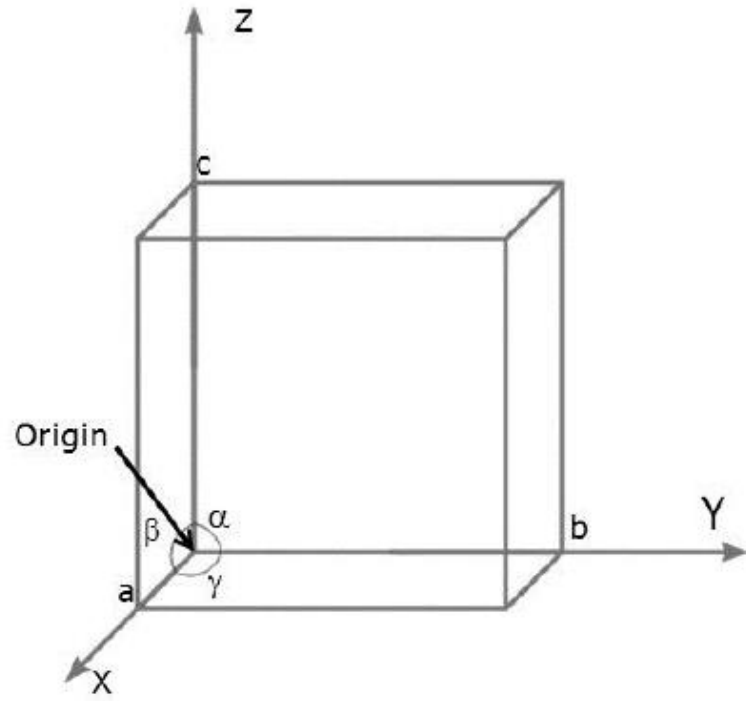


NaCl-type structure model of grain of salt magnified x 1 billion times

- Luckily, it is sufficient to describe the symmetry and shape of the repeat unit (**unit cell**) and its content (**asymmetric unit**).

Unit cell

A unit cell is the most basic building block of a crystal that contains the complete crystal pattern.

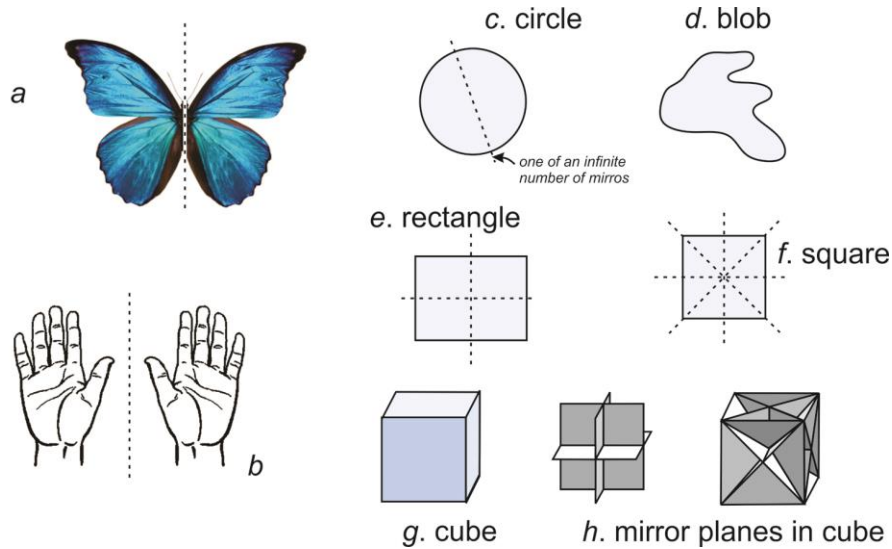


- Defined by three lengths (a , b , and c) referred to as **crystallographic axes**
- Angles between these axes are referred to as α , β , and γ
- Lattice parameters (a , b , c , α , β , γ) $\rightarrow$ dimensions of the unit cell

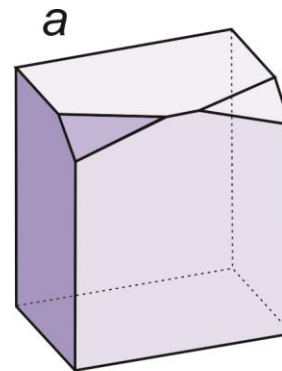
Symmetry

- The symmetry of a crystal is the geometrical relationship between its faces and edges.
- Crystal symmetry is a reflection of internal atomic symmetry.
- If a crystal has symmetry, the symmetry is common to all of its properties.
- Crystals may have any of an infinite number of shapes, but the number of possible symmetries is limited.
- Crystal symmetry is the basis for dividing crystals into different groups and classes.

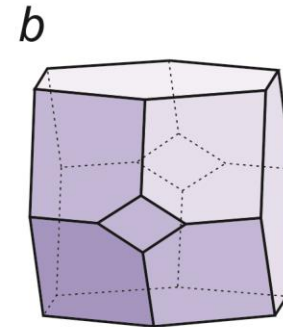
1- Mirror planes (m)



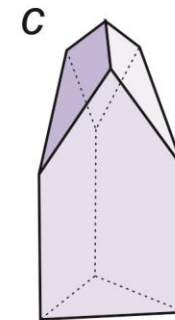
A face, or any object, on one side of a mirror has an equivalent at an equal distance on the other side of the mirror.



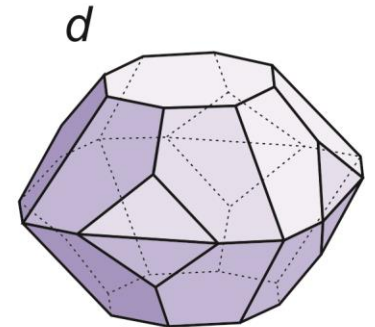
single mirror



three
perpendicular
mirrors

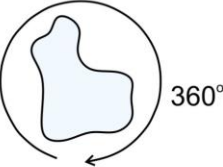
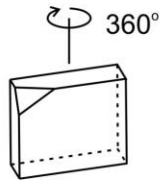
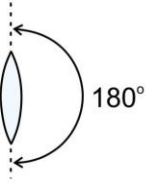
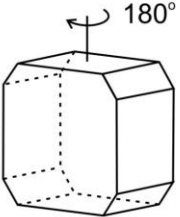
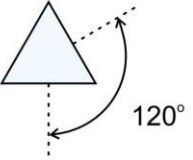
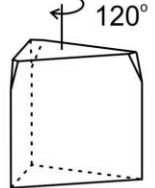
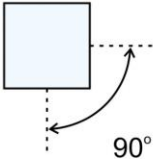
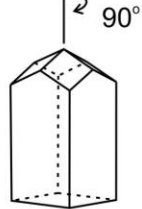
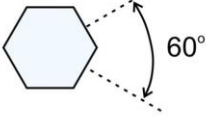
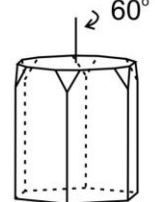


three mirrors at
120° to each
other



four vertical and
one horizontal
mirror

2- Rotational symmetry (1, 2, 3, 4, 6)

symmetry operator	2-dimensional example	3-dimensional example
1		
2		
3		
4		
6		

Symmetry with respect to a line called a *rotation axis*.

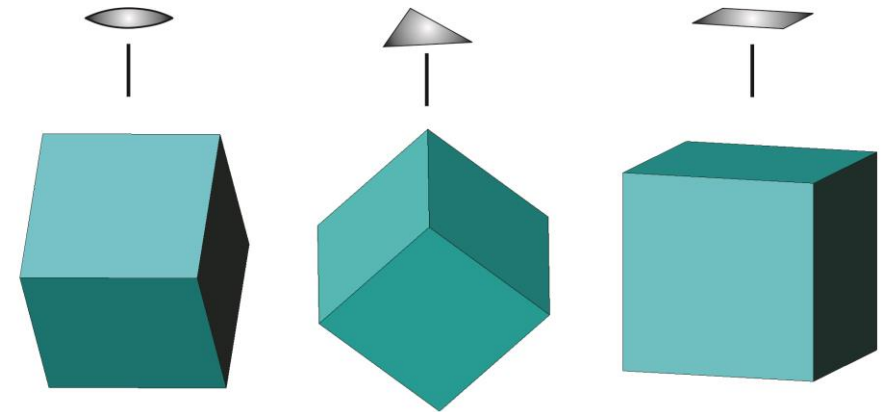
In two dimensions, a lens shape appears unchanged when rotated 180°. We say it has *2-fold symmetry* because two repeats of the rotation operation return it to its original position. A 2-fold rotation axis is perpendicular to the plane of the page.

3-fold

4-fold

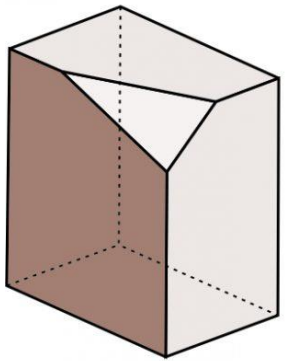
6-fold

$$\frac{2\pi}{n}$$

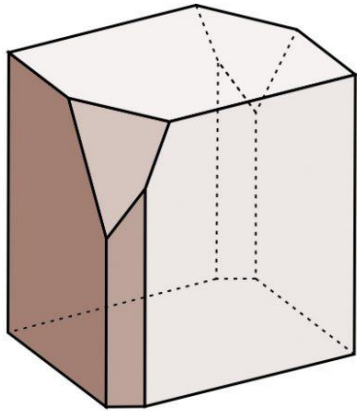


Rotational axes of a cube

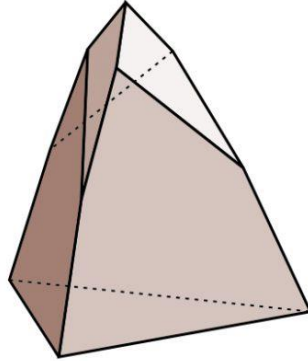
Rotational symmetry



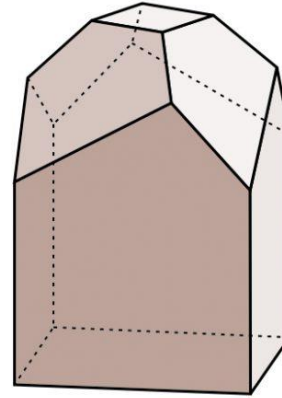
1-fold



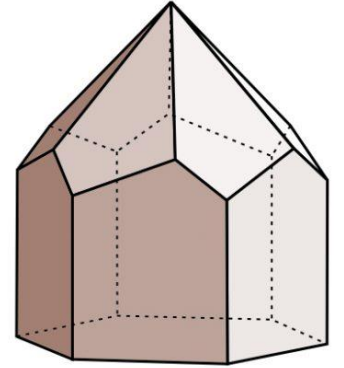
2-fold



3-fold

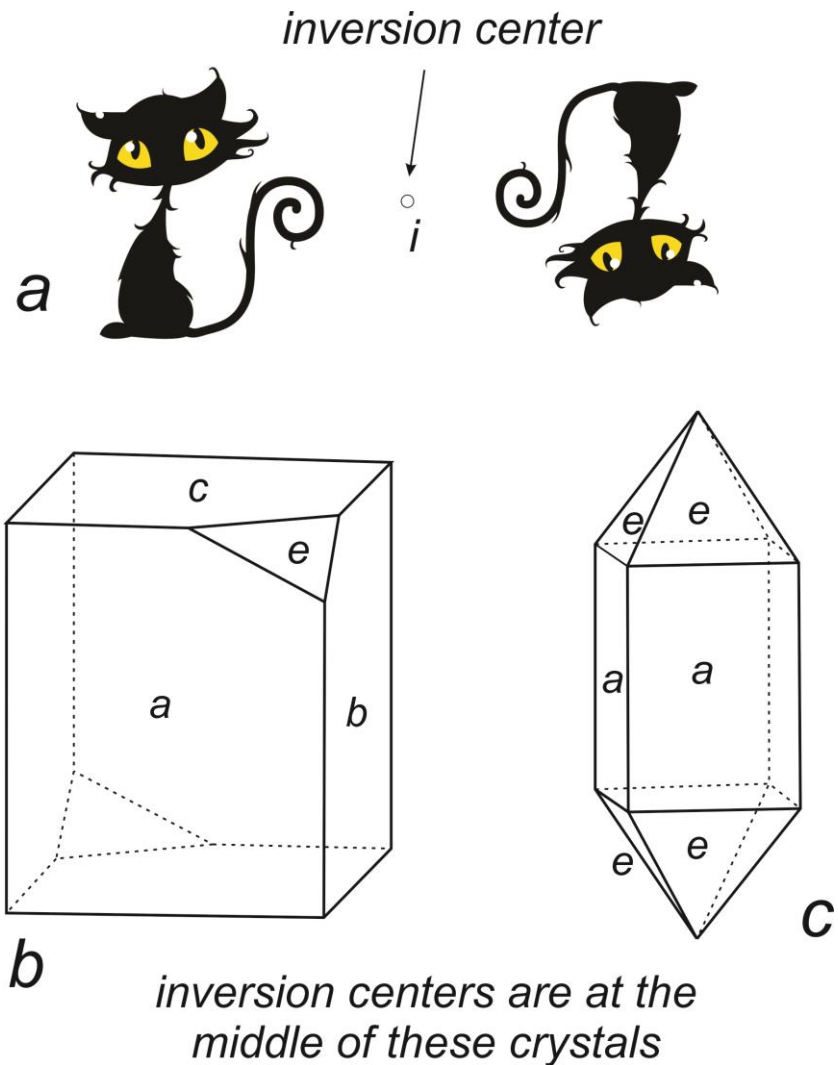


4-fold



6-fold

3- Inversion center ($\bar{1}$)



Symmetry with respect to a point.

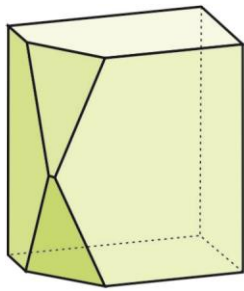
Inversion produces faces related in the same way that a lens may yield an upside-down and backward image. In two dimensions, inversion centers give the same results as 2-fold axes of symmetry.

In three dimensions, inversion symmetry is different from rotation. The crystal in figure b has an inversion center and no other symmetry.

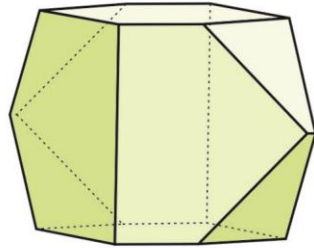
Figure c contains an inversion center, but other symmetry is present, too. 90° rotation relates the four *a*-faces. Inversion and rotation relate the eight *e*-faces. Additionally, many mirror planes are present.

Rotational with perpendicular mirror plane ($1/m, 2/m, 3/m, 4/m, 6/m$)

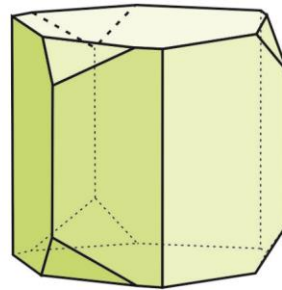
Many crystals have rotation axes that are perpendicular to mirror planes.



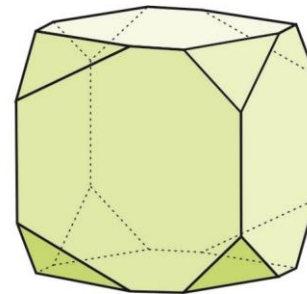
$$\frac{1}{m} = m$$



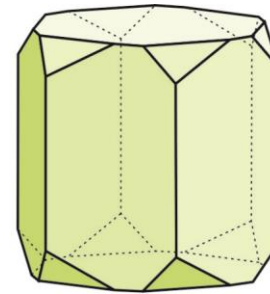
$$\frac{2}{m}$$



$$\frac{3}{m}$$



$$\frac{4}{m}$$



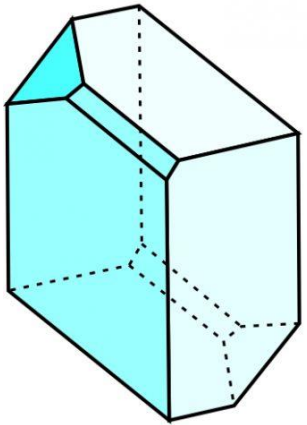
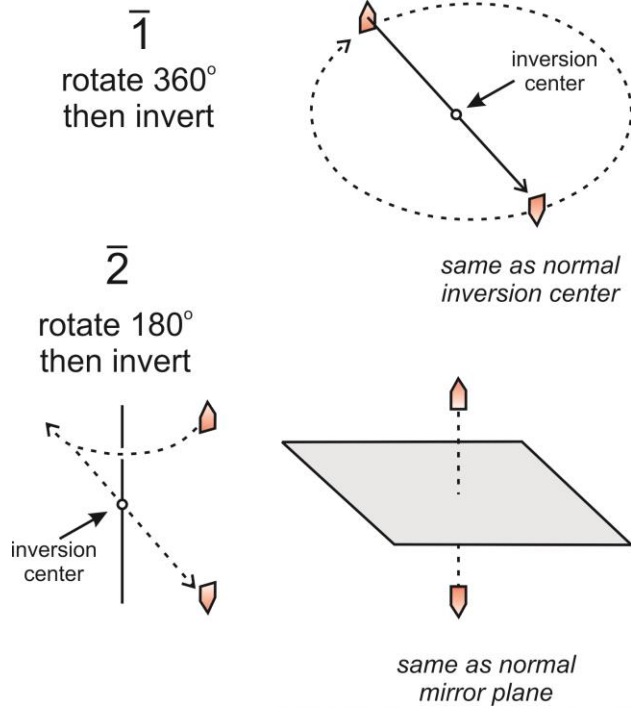
$$\frac{6}{m}$$

4- Rotoinversion ($\bar{n}$)

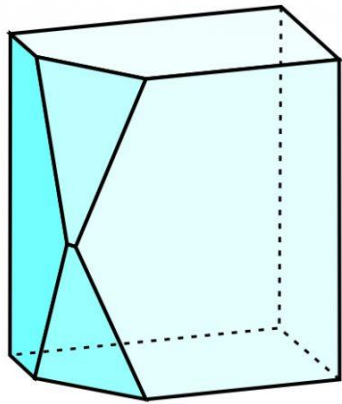
Rotoinversion, a combination of rotation and inversion, is a symmetry operation sometimes distinct from the others.

The symbols $\bar{1}$, $\bar{2}$, $\bar{3}$, $\bar{4}$, $\bar{6}$, represent rotoinversion axes.

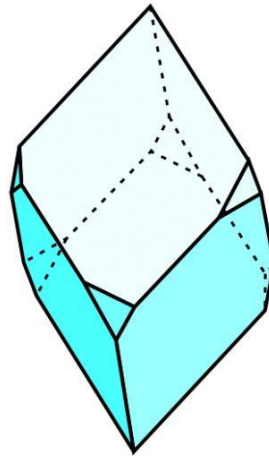
They are articulated as “bar-1,” “bar-2,” etc.



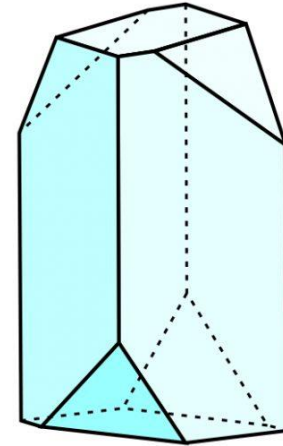
$\bar{1}$



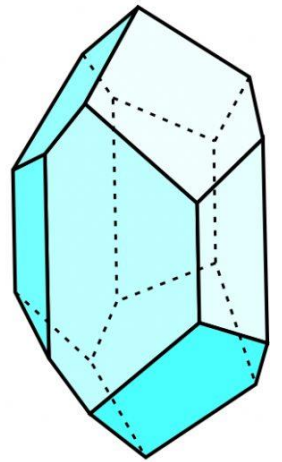
$\bar{2}$



$\bar{3}$



$\bar{4}$

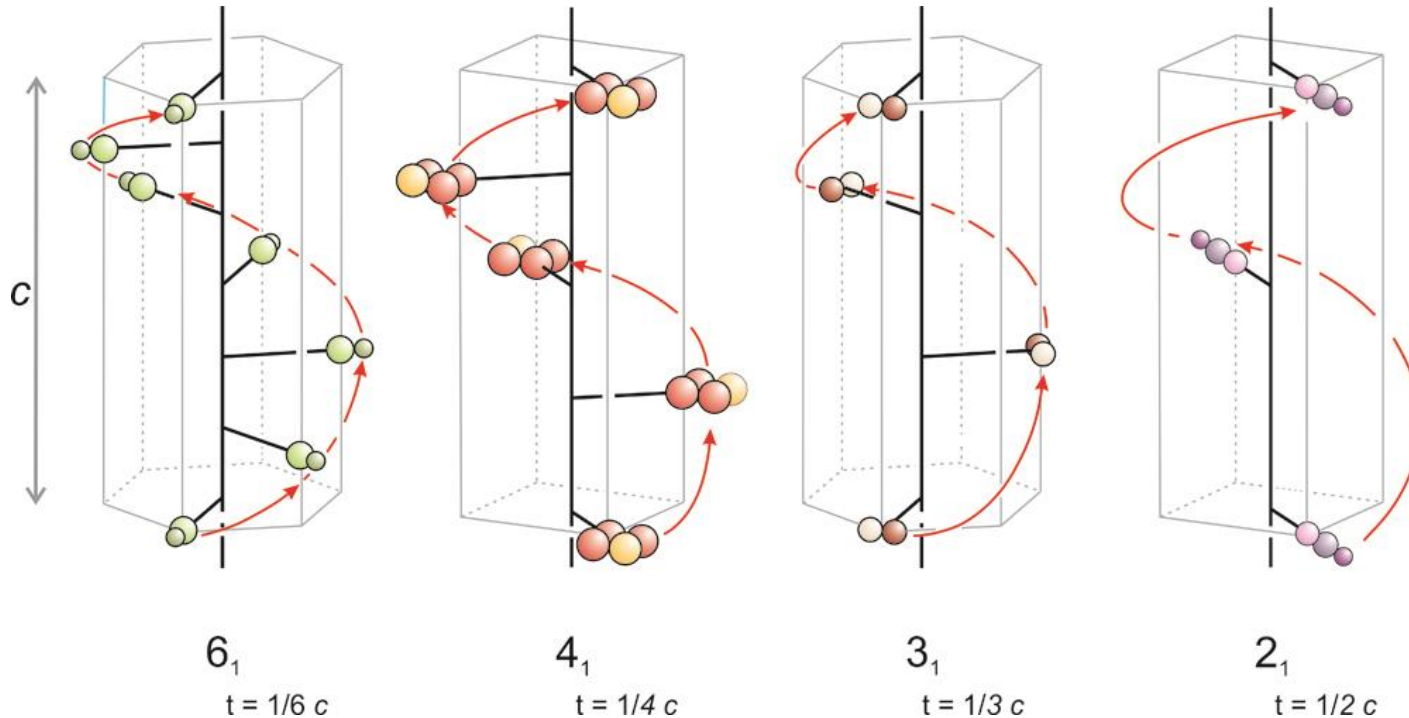


$\bar{6}$

Combination of symmetry

Combining the rotation axes and the mirror planes with the characteristic translations of the crystals, new symmetry elements appear, with some "sliding" components: *screw axes* (or *helical axes*) and *glide planes*.

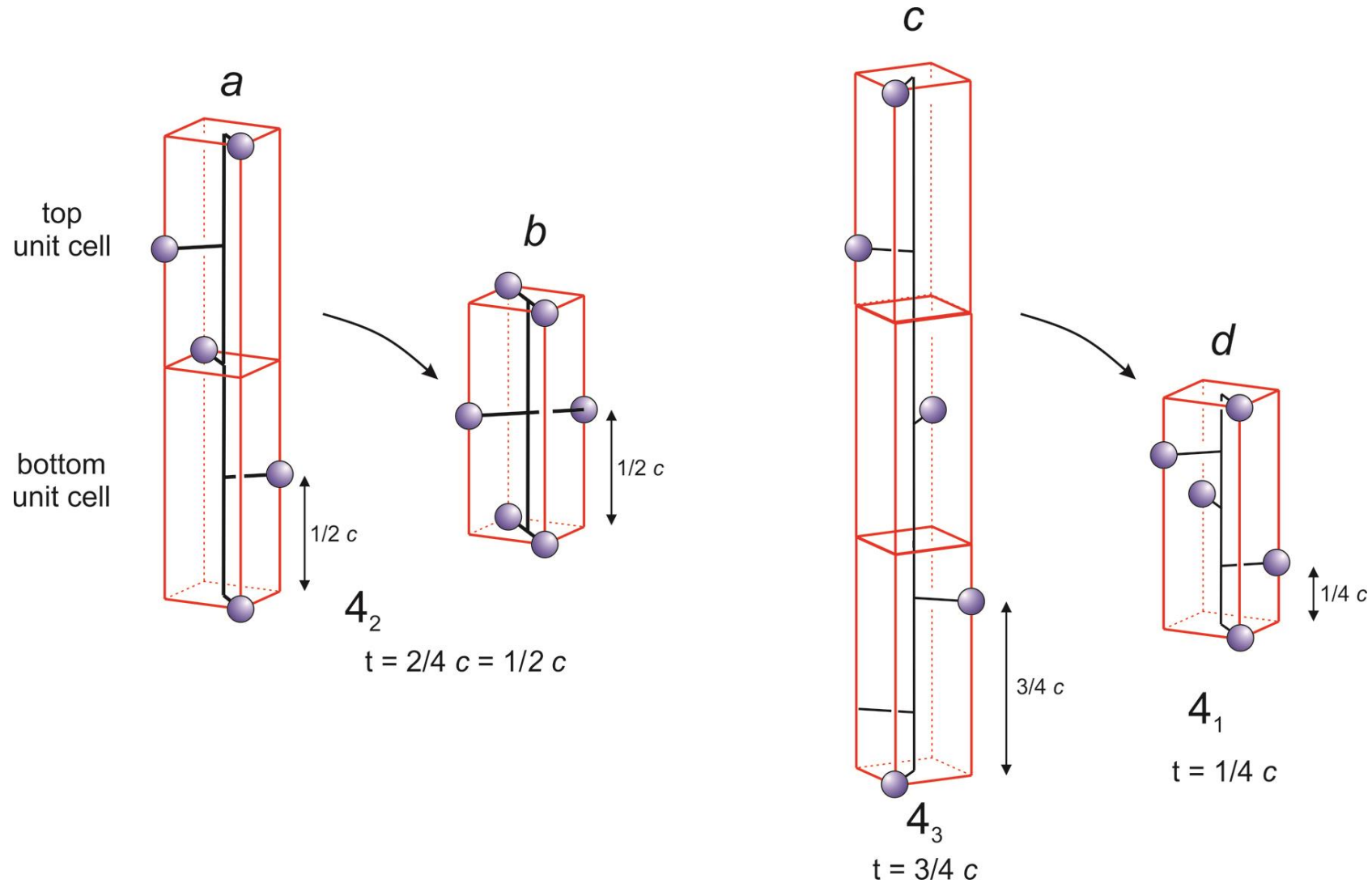
Rotation + translation = screw axes (elicogira)



A screw axis has the appearance of a spiral staircase. We rotate a motif, translate it, and get an additional motif. Each n -fold screw operation involves rotation of $360^\circ/n$. After n repeats, the screw has come full circle.

The translation associated with a screw axis must be a rational fraction of the unit cell dimension or the result will be an infinite number of atoms, all in different places in different unit cells.

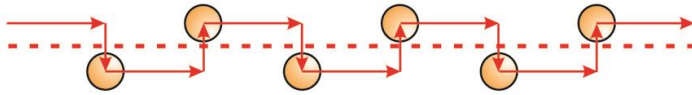
Rotation + translation = screw axes (elicogira)



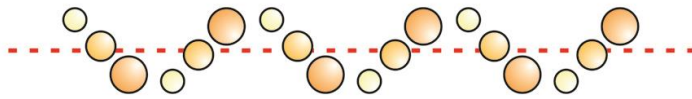
Reflection + translation = glide plane (slittopiano)

Glide planes differ from normal mirror planes because they involve translation before reflection.

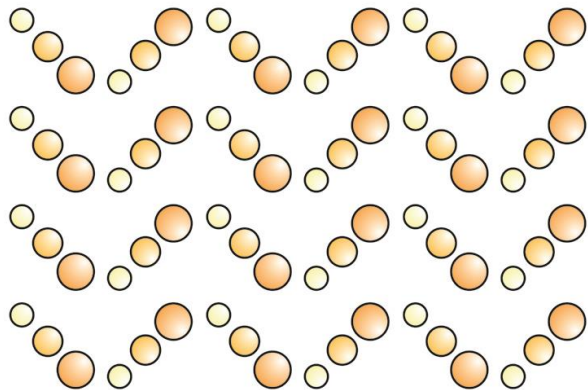
a. single atom repeating by a glide plane



b. three atom motif repeating by a glide plane



c. 2-d collection of atoms with glide plane symmetry

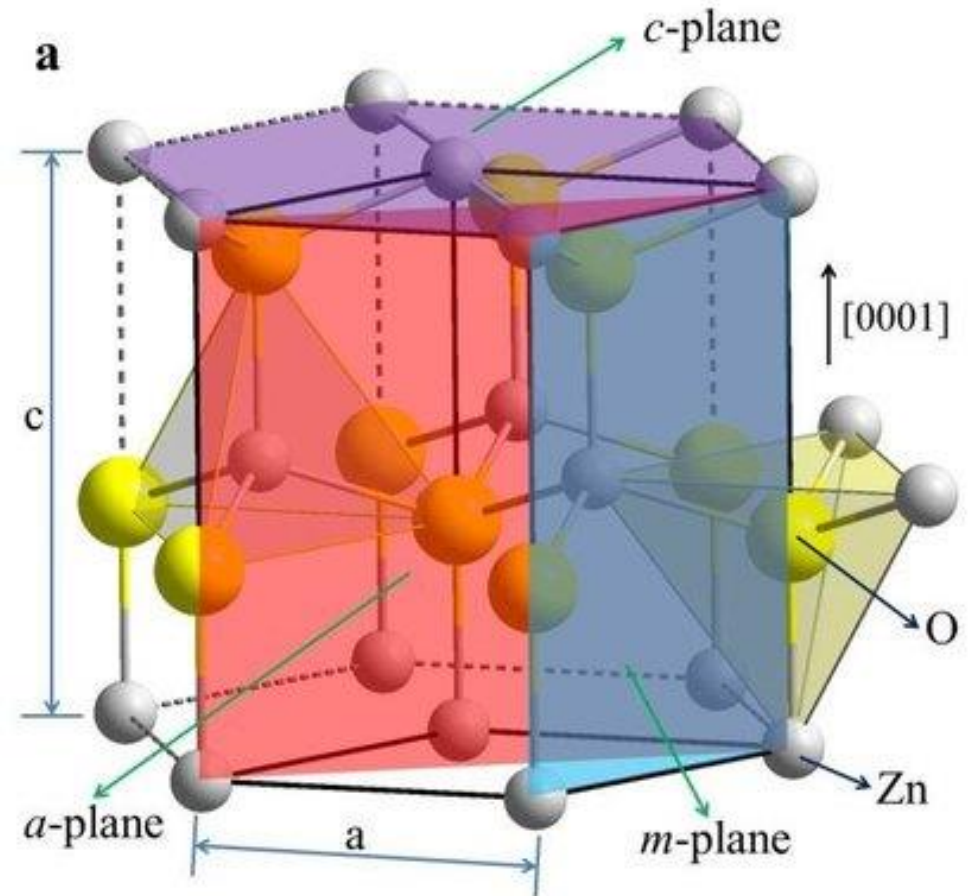
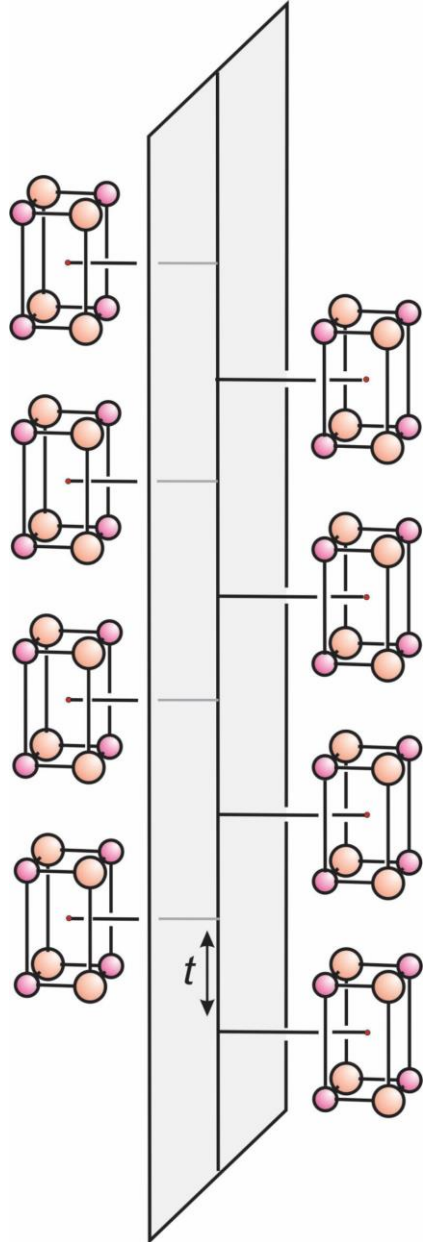


a- a single atom is repeated by a horizontal glide plane (dashed red line). The red arrows show how the atom repeats by translation followed by reflection.

b- a three-atom motif repeats according to a horizontal glide plane.

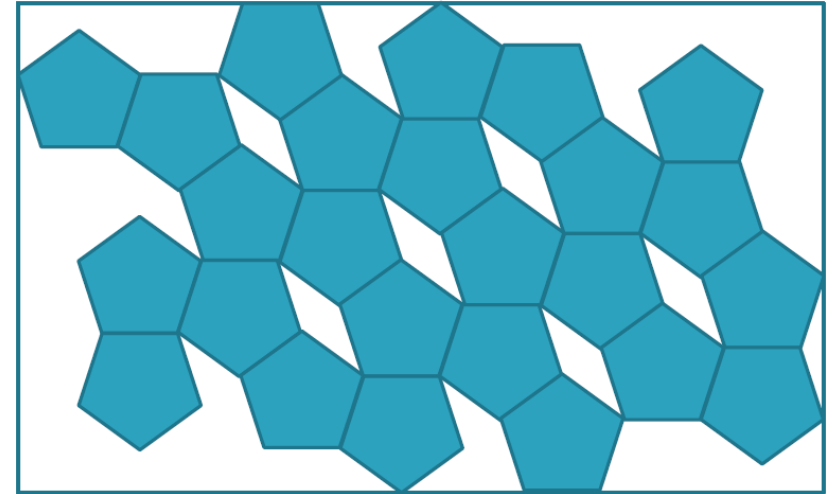
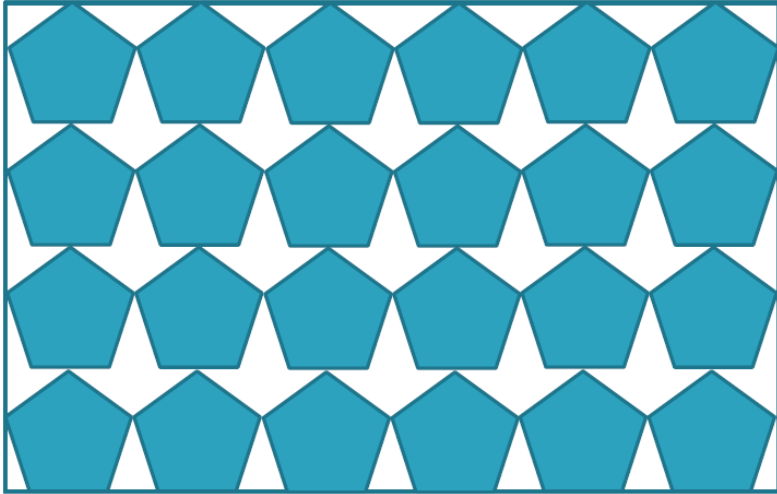
c- a 2D pattern with glide plane symmetry. The pattern contains no mirrors, rotation axes, and no inversion center.

Reflection + translation = glide plane (slittopiano)



Wurtzite (ZnO)

Forbidden symmetry in a crystal



A 5-fold axis of symmetry cannot exist in a periodic lattice because it is not possible to find an arrangement where there will be no gaps. It is impossible to fill all the area of a plane with a connected array of pentagons.

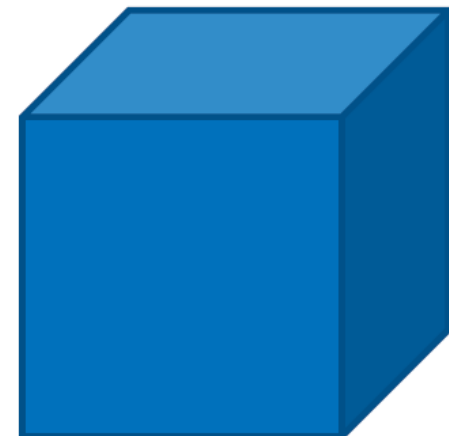
Rotation:

$$\frac{2\pi}{n} = 1, 2, 3, 4, 6$$

Crystal systems

Crystal systems	Axes system
Cubic	$a = b = c , \alpha = \beta = \gamma = 90^\circ$
Tetragonal	$a = b \neq c , \alpha = \beta = \gamma = 90^\circ$
Hexagonal	$a = b \neq c , \alpha = \beta = 90^\circ , \gamma = 120^\circ$
Rhomboedric	$a = b = c , \alpha = \beta = \gamma \neq 90^\circ$
Orthorhombic	$a \neq b \neq c , \alpha = \beta = \gamma = 90^\circ$
Monoclinic	$a \neq b \neq c , \alpha = \gamma = 90^\circ , \beta \neq 90^\circ$
Triclinic	$a \neq b \neq c , \alpha \neq \gamma \neq \beta$

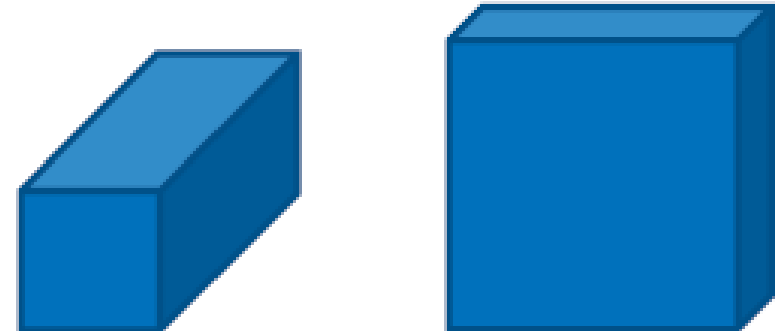
“Shape” of unit cell → crystal system



Crystal systems

Crystal systems	Axes system
Cubic	$a = b = c, \alpha = \beta = \gamma = 90^\circ$
Tetragonal	$a = b \neq c, \alpha = \beta = \gamma = 90^\circ$
Hexagonal	$a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$
Rhomboedric	$a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
Orthorhombic	$a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$
Monoclinic	$a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
Triclinic	$a \neq b \neq c, \alpha \neq \gamma \neq \beta$

“Shape” of unit cell → crystal system



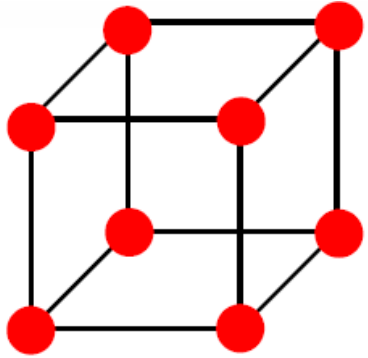
Crystal systems

Crystal systems	Axes system
Cubic	$a = b = c, \alpha = \beta = \gamma = 90^\circ$
Tetragonal	$a = b \neq c, \alpha = \beta = \gamma = 90^\circ$
Hexagonal	$a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$
Rhomboedric	$a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
Orthorhombic	$a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$
Monoclinic	$a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
Triclinic	$a \neq b \neq c, \alpha \neq \gamma \neq \beta$

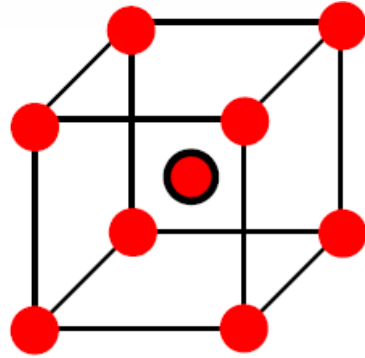
“Shape” of unit cell → crystal system



Bravais lattices



Primitive (P)



- Within each crystal system, different types of centering produce a total of 14 different Bravais lattices.

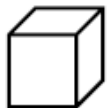
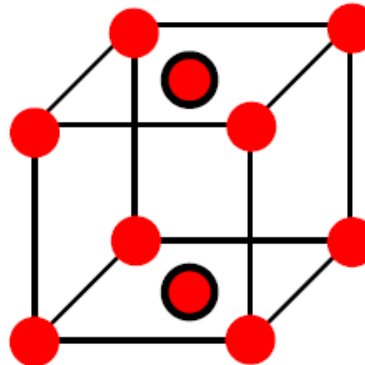
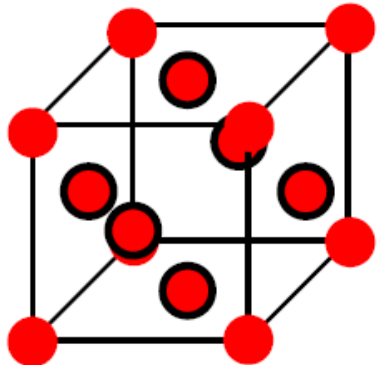
- P – Simple (Primitive)

- I – Body Centered

- F – Face Centered

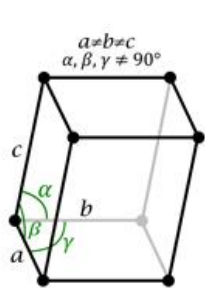
- C – Base Centered

- All crystalline materials can have their crystal structure described by one of these Bravais lattices.

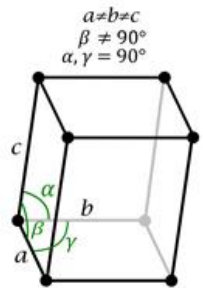


Unit cell

● Lattice point

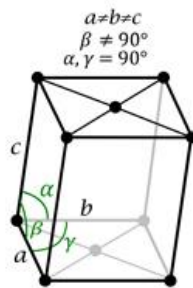


Triclinic

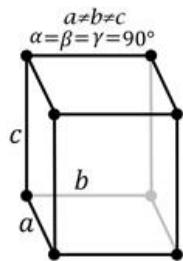


P

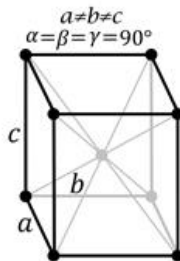
Monoclinic



C

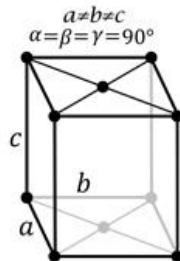


P

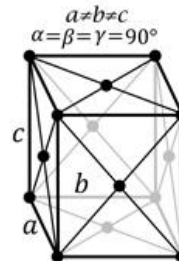


I

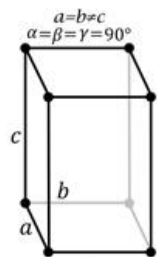
Orthorhombic



C

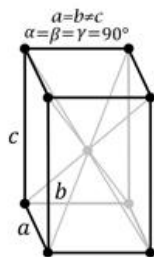


F

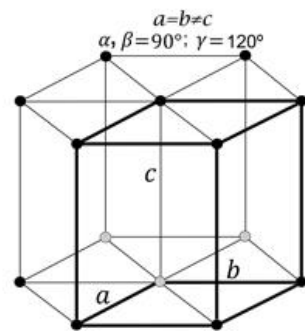


P

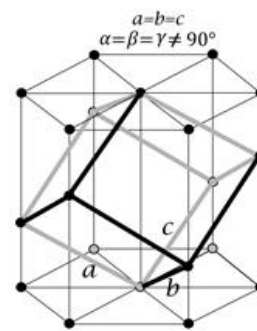
Tetragonal



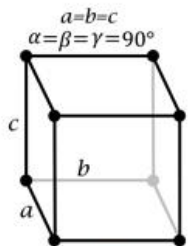
I



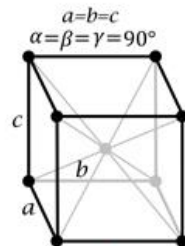
Trigonal / Hexagonal P



Trigonal R

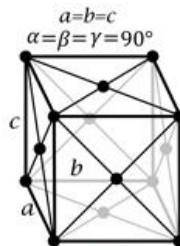


P



I

Cubic



F

7 crystal classes (systems)

14 Bravais lattices

Point groups

In 3 dimensions, there are 32 point groups. Point groups are mathematical constructs that capture all the non-translation symmetry options that can be performed on an object (reflection, rotation, or rotoinversion).

- Fit each point group to a crystal system: Triclinic, monoclinic, orthorhombic, tetragonal, trigonal, hexagonal, and cubic.
- Write the highest-order symmetry on each perpendicular axis. The “/” sign indicates that 2 operations are present in the same axis.

$6/mmm \rightarrow \underline{6/m} \underline{m} \underline{m}$

- 6-fold rotation + mirror plane in the x-direction
- mirror plane in the y-direction
- mirror plane in the z-direction.

Crystal System	Number of Point Groups	Crystallographic Point Groups
Cubic	5	$23 \bar{m} \bar{3} 432$ $43m \ m\bar{3}m$
Hexagonal	7	$6 \bar{6} 6/m \ 622$ $6mm \ \bar{6}2m \ 6/mmm$
Trigonal	5	$3 \bar{3} 32 \ 3m \ \bar{3}m$
Tetragonal	7	$4 \bar{4} 4/m \ 422$ $4mm \ \bar{4}2m \ 4/mmm$
Orthorhombic	3	$222 \ mm2 \ mmm$
Monoclinic	3	$2 \ m \ 2/m$
Triclinic	2	$1 \ \bar{1}$

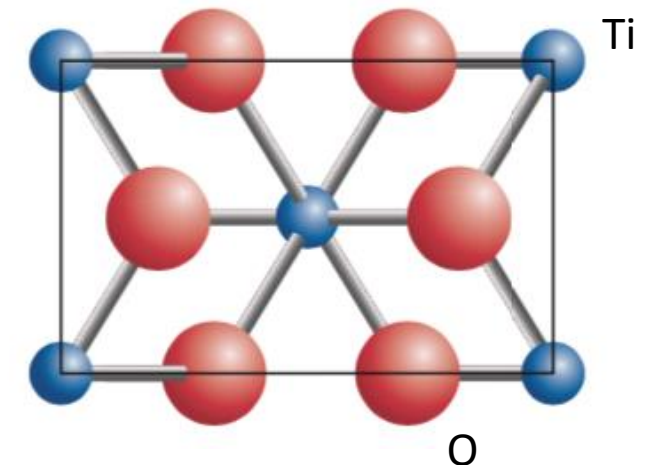
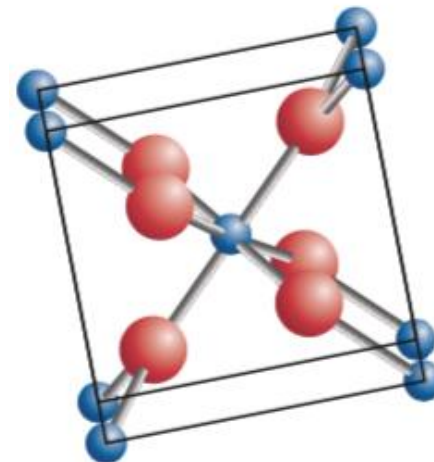
32 crystallographic point groups of different crystal systems.

Space groups

- When we combine the symmetry operators with the 14 possible space lattices, we get 230 possible space groups.
- Deriving them all is not trivial, and crystallographers debated the exact number until the 1890s when several independent studies concluded that there could only be 230.
- Crystallographers use several different notations for space groups; the least complicated is that used in the *International Tables for X-ray Crystallography* (ITX) (Hahn, 1983). ITX space group symbols consist of a letter indicating lattice type (*P*, *I*, *F*, *C*) followed by symmetry notation.

Example: $P4_2/mnm$ (136), the space group of rutile.

- Primitive (*P*) tetragonal unit cell
- 4_2 screw axis perpendicular to a mirror plane
- *n* glide plane
- *m* mirror plane



Nomenclature

Symmetry elements

Symmetry Elements

there are **5 types** in point symmetry

- | | | |
|--|--------------|-----------|
| 1. center of symmetry (or inversion): | point | $\bar{1}$ |
| 2. rotation (or proper) axis | : line | n |
| 3. mirror | : plane | m |
| 4. rotation–inversion axis | : line | $\bar{n}$ |
| 5. identity | : no element | 1 |

Translational Symmetry

in **repeating** lattices, two **additional** symmetry elements

translational elements

1. **screw axis** rotation **and** translation: n_r
rotation by $360^\circ/n$;
followed by translation of r/n along **that axis** (**a**, **b** or **c**)
2-fold screw axis most common: 2_1
2. **glide plane** reflection **and** translation: a, b, c, n or d
reflection across plane;
followed by translation of $1/2$ (usually) unit cell parallel to
plane along **a**, **b**, **c**, **face diagonal** (n), or **body diagonal** (d)

Space Group Nomenclature

space group name comes from Bravais lattice symbol, modified for translational symmetry

easy to understand the components of **many** names, especially **monoclinic** and **orthorhombic**:

P2₁/c (P 2-1 on c)

primitive unit cell (1 lattice point)

2-fold screw axis along **b** (**unique axis**)

c glide (translation along **c** axis) in **ac** plane ($\perp$ to **b**)

Pbca **primitive** unit cell (1 lattice point)

b glide (translation along **b** axis) in **bc** plane ($\perp$ to **a**)

c glide (translation along **c** axis) in **ac** plane ($\perp$ to **b**)

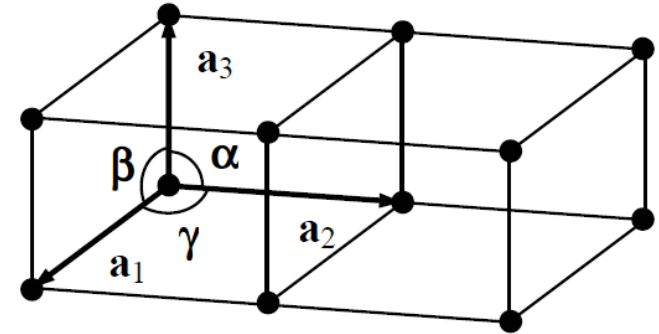
a glide (translation along **a** axis) in **ab** plane ($\perp$ to **c**)

Crystal lattice

- The lattice is defined by fundamental translation vectors. For example, the position vector of any lattice site of the can be written as:

$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3$$

$\vec{a}_1, \vec{a}_2, \vec{a}_3$ primitive translation vectors,
 n_1, n_2, n_3 triplet of integers whose values depend on a particular lattice site.

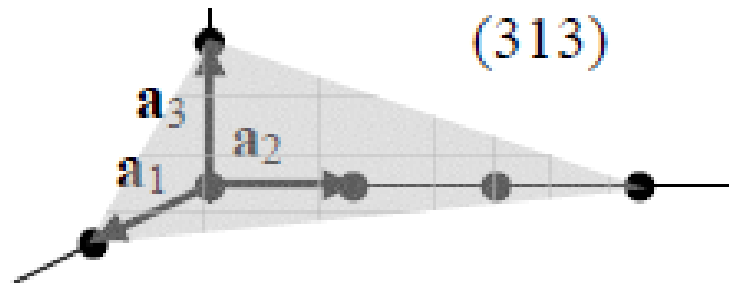
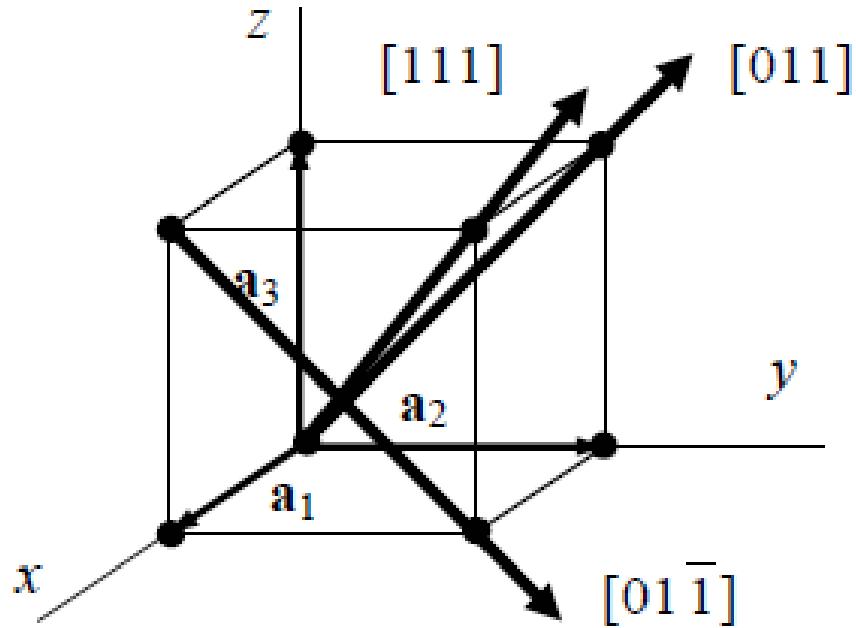


The volume of the unit cell represented by a parallelepiped with sides $\vec{a}_1, \vec{a}_2, \vec{a}_3$ is given by

$$V = |\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3|$$

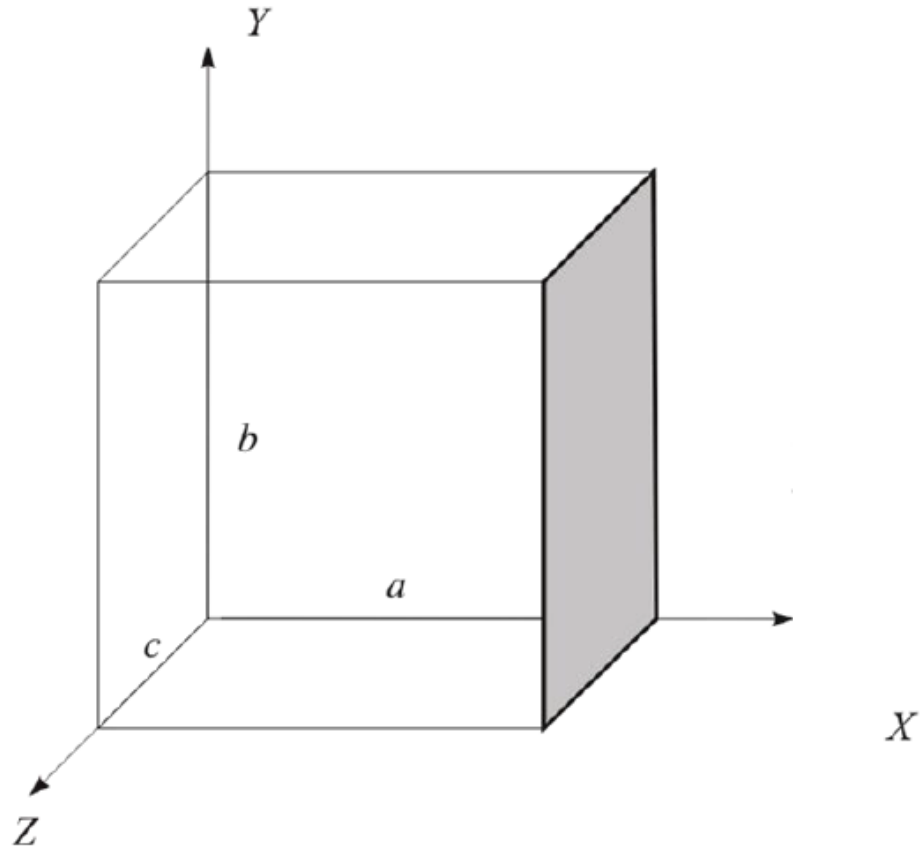
It is sometimes more convenient to deal with non-primitive or conventional cells, which have additional lattice sites either inside the cell or on its surface.

Crystal directions



- Any lattice vector can be written as:
$$\vec{R} = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3$$
- The direction is specified by the three integers n_1, n_2, n_3
- About directions, we mean a whole set of parallel lines, which are equivalent due to translational symmetry
- Lattice planes are an abstract but useful concept to describe the crystal structure
- Sets of parallel, evenly spaced planes extended throughout the crystal
- The distance d between the planes is probed by Bragg's law
- Method for identifying diffracting planes in a crystal

Miller Indexes



➤ Intersection of the plane along X, Y, Z are:

$(1, \infty, \infty)$ Weiss indexes

➤ Take the inverse:

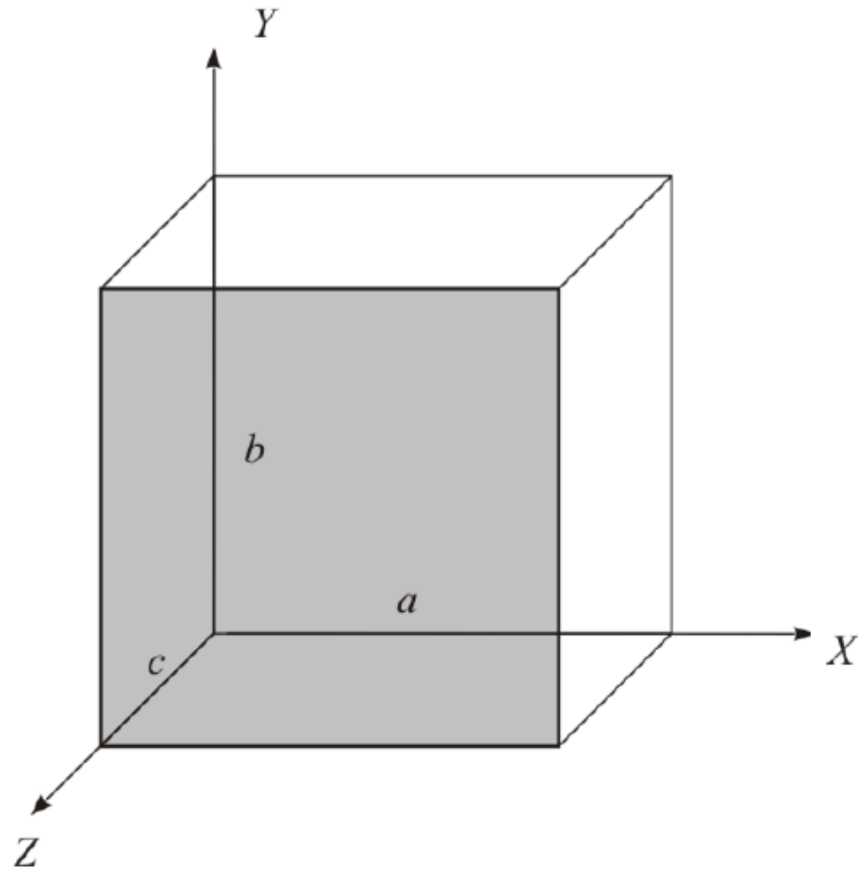
$(1\ 0\ 0)$ Miller indexes

This set is called Miller indices of the plane (hkl)

Direction [hkl]

Plane (hkl)

Miller Indexes

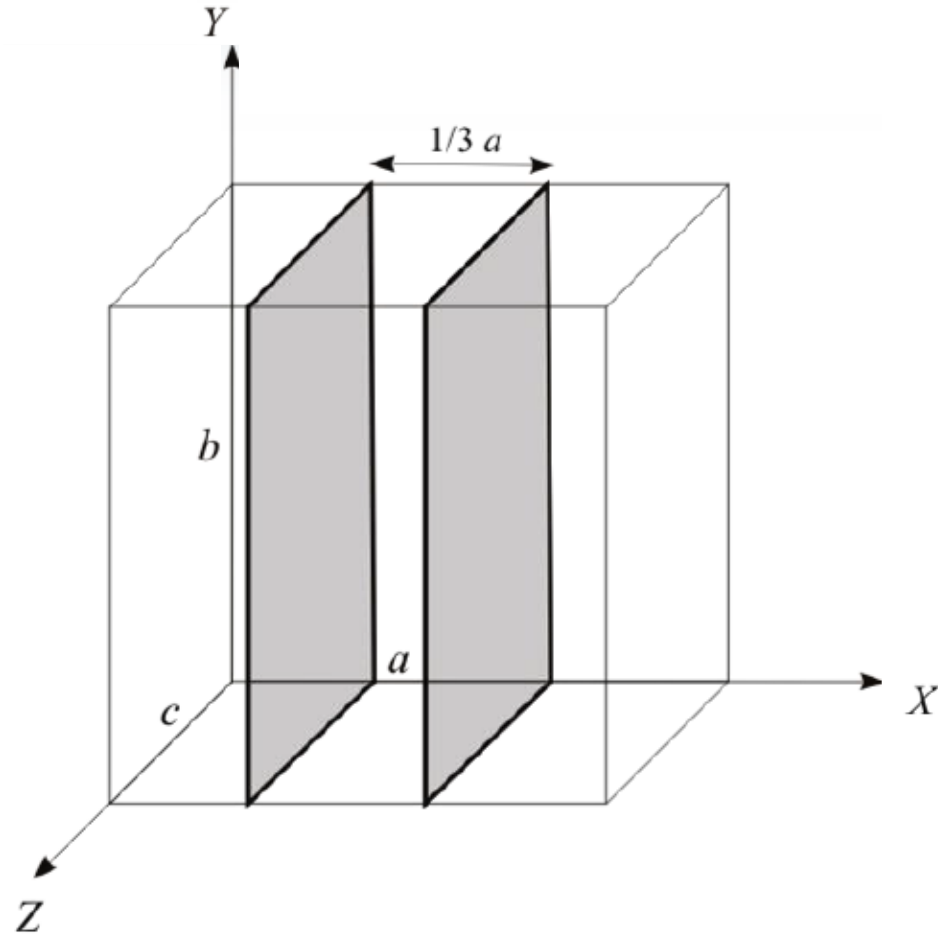


➤ Intersection of the plane along X, Y, Z are:

$(\infty, \infty, 1)$ Weiss indexes

$(0\ 0\ 1)$ Miller indexes

Miller Indexes



➤ Intersection of the plane along X, Y, Z are:

$(1/3, \infty, \infty)$ Weiss indexes

$(3\ 0\ 0)$ Miller indexes

For the second plane $(2/3, \infty, \infty)$

Miller Indexes should be integers, so removing fractions we have $(3\ 0\ 0)$

Wyckoff sites

- Where the atoms in a crystal can be found.
- Number (multiplicity) and letter (Wyckoff site with no physical meaning).
- Multiplicity: how many atoms are generated by symmetry if we place a single atom at that position.

```
Lattice type      P
Space group name  P m -3 m
Space group number 221
Setting number    1

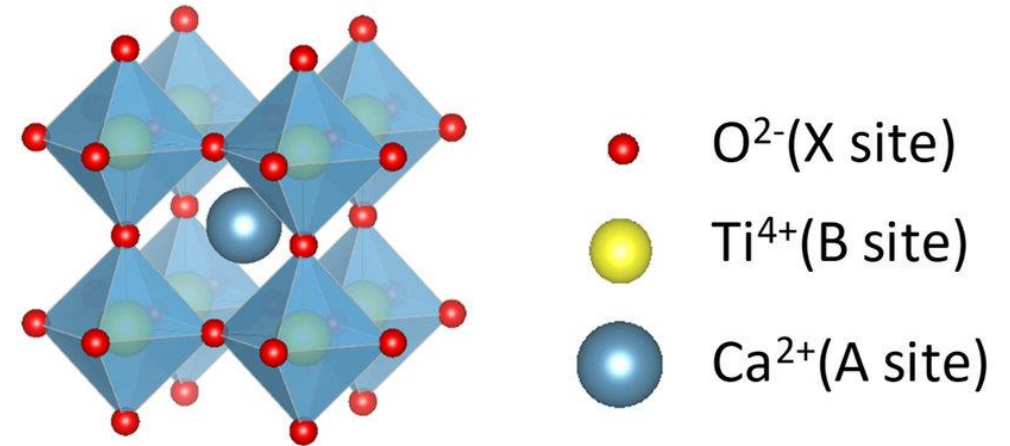
Lattice parameters

  a      b      c    alpha    beta    gamma
 3.79500  3.79500  3.79500  90.0000  90.0000  90.0000
```

Unit-cell volume = 54.655688 Å³

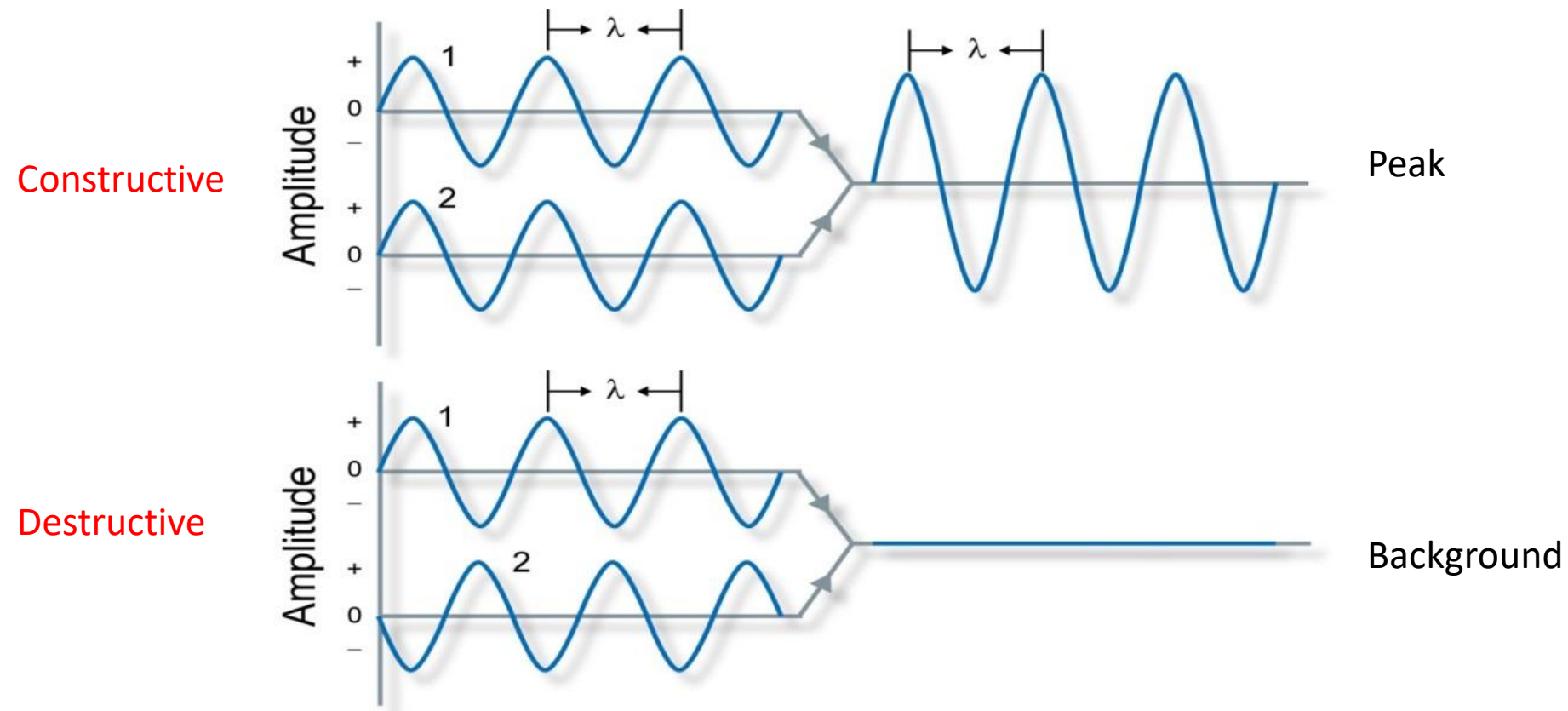
Structure parameters

			x	y	z	Occ.	U	Site	Sym.
1	Ca	Ca1	0.00000	0.00000	0.00000	1.000	0.000	1a	m-3m
2	Ti	Ti1	0.50000	0.50000	0.50000	1.000	0.000	1b	m-3m
3	O	O1	0.50000	0.50000	0.00000	1.000	0.000	3c	4/mmm. m



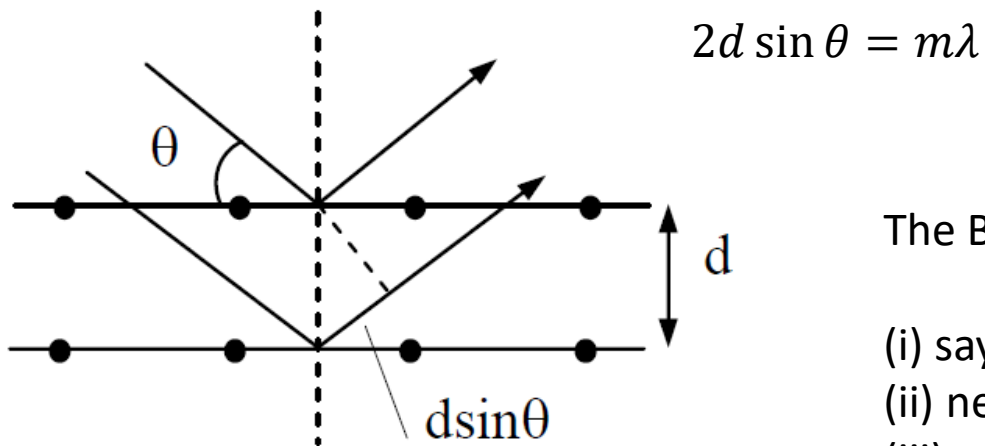
X-ray diffraction

- Diffraction occurs when light (waves) is scattered by a periodic array with long-range order (lattice), producing constructive interference at specific angles (diffraction pattern)



Bragg's law

- In 1913 Bragg found that crystalline solids have remarkably characteristic patterns of reflected x-ray radiation. In crystalline materials, for certain wavelengths and incident directions, intense peaks of scattered radiation were observed.
- Bragg accounted for this by regarding a crystal as made out of parallel planes of atoms, spaced by distance d apart. The conditions for a sharp peak in the intensity of the scattered radiation were that:
 - (1) the x-rays should be specularly reflected by the atoms in one plane;
 - (2) the reflected rays from the successive planes interfere constructively



The Bragg law is greatly oversimplified (but it works!). It

- says nothing about **intensity** and **width** of x-ray diffraction peaks;
- neglects differences in scattering from **different atoms**;
- neglects **distribution of charge** around atoms.

Reciprocal lattice

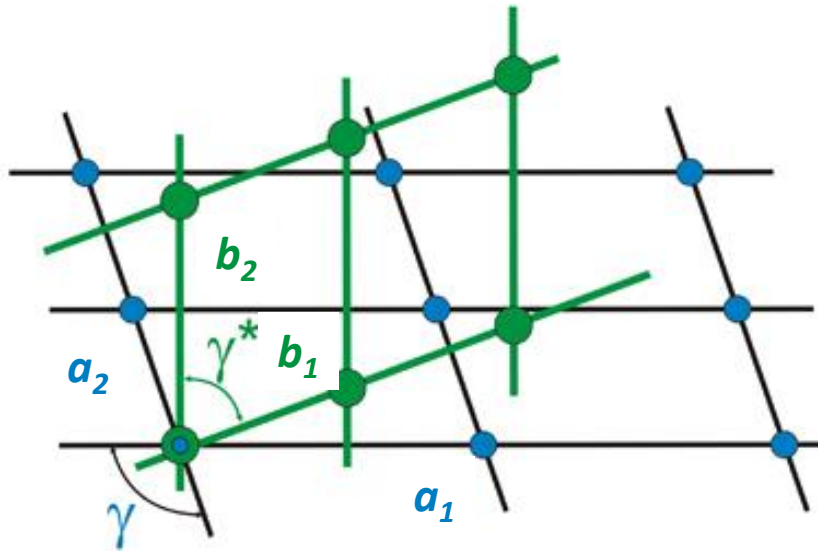
- To describe the morphology of a crystal we can associate, with each set of parallel lattice planes, a vector drawn from a given origin and normal to the corresponding lattice planes.
- Each vector has a length directly related to the spacing of the lattice planes

Why is the reciprocal lattice important?

- The nature of the diffraction pattern is governed by the periodicity and the positions of the diffraction spots depend directly on the properties of the lattice.
- This operation transforms the direct space into an associate space, the reciprocal space, and the diffraction spots of a crystal are associated with the nodes of its reciprocal lattice

Reciprocal lattice

- A crystal is composed of identical atoms placed at the lattice sites $\mathbf{R}$ and each atom can reradiate the incident radiation in all directions.
- Sharp peaks are observed only in the directions and at wavelengths for which the x-rays scattered from all lattice points interfere constructively.



$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3 \quad \text{crystal lattice point}$$

$$\vec{G} = h_1 \vec{b}_1 + m_2 \vec{b}_2 + m_3 \vec{b}_3 \quad \text{reciprocal lattice point}$$

$$\vec{b}_1 = \frac{2\pi}{V} \vec{a}_2 \times \vec{a}_3$$

$$\vec{b}_2 = \frac{2\pi}{V} \vec{a}_3 \times \vec{a}_1$$

$$\vec{b}_3 = \frac{2\pi}{V} \vec{a}_1 \times \vec{a}_2$$

$$\vec{a}_i \cdot \vec{b}_j = 2\pi \delta_{ij}$$

or

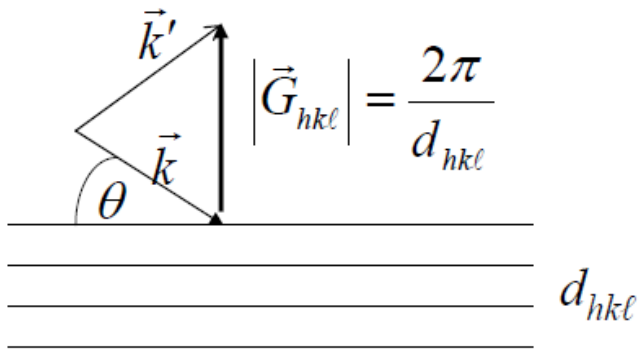
$$\vec{R} \cdot \vec{G} = 2\pi m$$

Diffraction condition and reciprocal lattice

Two scattering atoms separated by a lattice vector $\vec{R}$ (elastic scattering).

- Let X-rays are incident from infinity, along direction $\hat{k}$ with wavelength λ and wavevector $\vec{k} = 2\pi/\lambda$.
- The path difference between the X-ray scattered from the two atoms should be $m\lambda$

$$(\hat{k}' - \hat{k}) \cdot \vec{R} = m\lambda \quad \text{constructive interference} \quad (\vec{k}' - \vec{k}) \cdot \vec{R} = 2\pi m$$



Defining the scattering wave vector $\Delta\vec{k} = \vec{k}' - \vec{k}$, the diffraction condition can be written as:

$$\Delta\vec{k} = \vec{G} \quad \text{where} \quad \vec{R} \cdot \vec{G} = 2\pi m$$

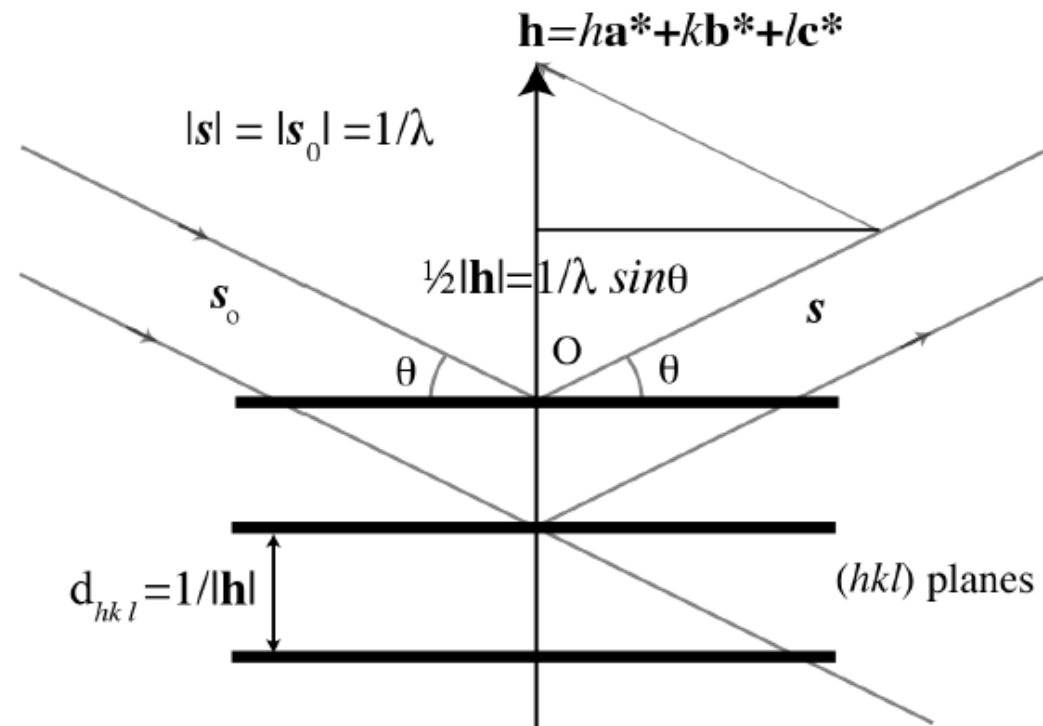
Laue condition

Laue diffraction condition and Bragg's equation

There is a direct relationship between the Laue condition of diffraction and the Bragg's equation.

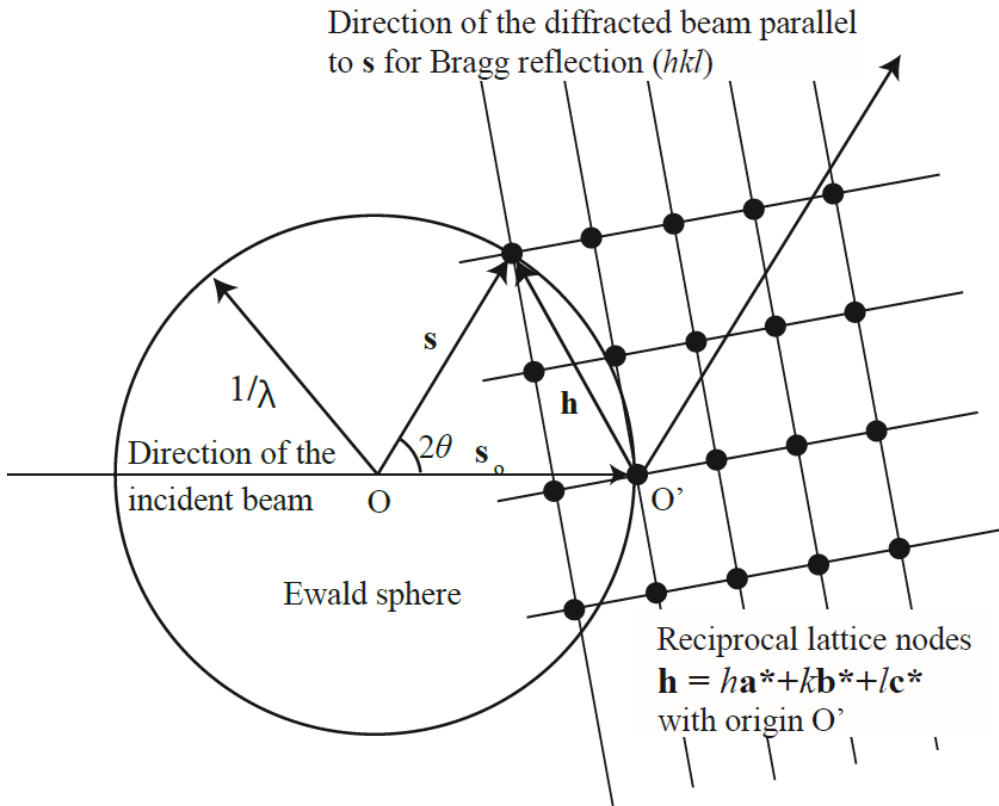
Laue: vector equation (relation between the reciprocal lattice vectors $\mathbf{h}$ and the incident $\mathbf{s}_0$ and diffracted beam $\mathbf{s}$)

Bragg: purely scalar



Ewald's sphere

The Ewald sphere is a very useful tool to determine the direction of the diffracted beam relative to the incident beam for a specific reciprocal lattice vector $\mathbf{h}$.



The Ewald sphere is based on the Laue condition for diffraction:

$$\mathbf{s} - \mathbf{s}_0 = \mathbf{h} \quad \mathbf{h} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^* \quad \text{reciprocal lattice vector}$$

The Ewald sphere is represented below in form of a two-dimensional drawing with radius $1/\lambda$.

To bring one reciprocal lattice vector in diffraction position, it is necessary to rotate the crystal around O' so that the end of the $\mathbf{h}$ vector intersects the surface of the sphere.

All vectors $\mathbf{h}$ such that $|\mathbf{h}| \leq 2/\lambda$ can be brought into the diffraction position by a rotation.

Structure factor

- The structure factor sums the result of scattering from all of the atoms in the unit cell (f_j) to form a diffraction peak from the (hkl) planes of atoms.
- The intensity of Bragg peak is proportional to F_{hkl}^2 .

$$I_{hkl} \propto |F_{hkl}| \qquad F_{hkl} = \sum_{j=1}^m N_j f_j e^{2\pi i(hx_j + ky_j + lz_j)} \qquad f_j: \text{Fourier transform of } \rho(r)$$

x_j, y_j, z_j : position of atoms in the atomic planes (where the atoms are on the atomic planes)

f_j : **atomic scattering factor** (fattore di forma atomico). It depends on nature of atom, direction of scattering, and X-ray wavelength and quantifies the relative efficiency of scattering at any angle by the group of electrons in each atom

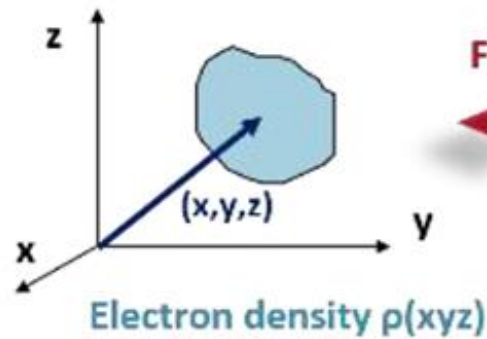
$$f \rightarrow Z \text{ for } \theta \rightarrow 0^\circ$$

N_j : fraction of every equivalent position occupied by atom j. Often an atom does not occupy every unit cell of the crystallite. This is the probability of the atom occupying position xyz

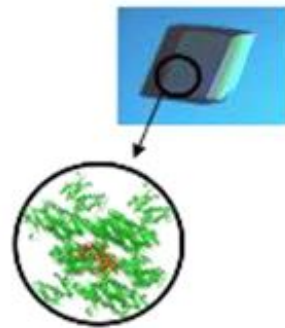
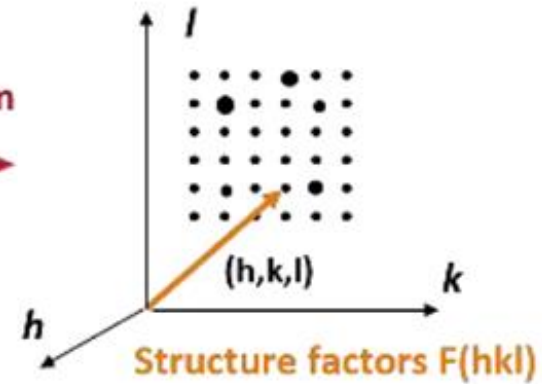
From crystals to diffraction patterns

$$\rho(xyz) = \frac{1}{V} \sum_{hkl} |F(hkl)| \cdot e^{-2\pi i[hx+ky+lz-\phi(hkl)]}$$

$$F(hkl) = \sum_j f_j e^{2\pi i(hx_j+ky_j+lz_j)}$$

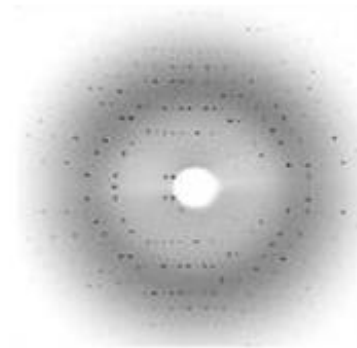


Fourier Transform
diffraction



Crystal
3D lattice in direct space

Fourier Transform
diffraction



Diffraction pattern
3D lattice in reciprocal space

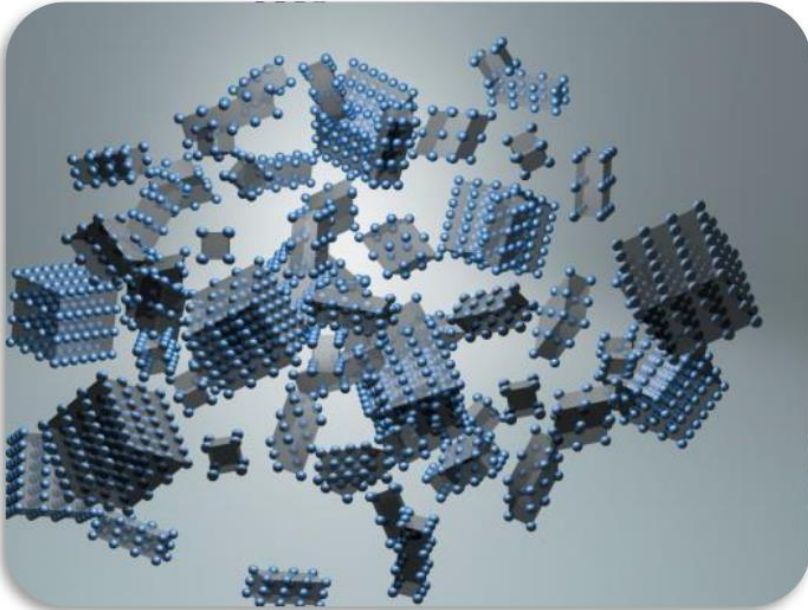
Extinction conditions

Depending on the structure, there may be some systematic extinctions

Bravais Lattice	Example Compounds/Structures	Allowed Reflections	Forbidden Reflections
Primitive (<i>P</i>) Cubic	Simple Cubic (α -Po)	Any h,k,l	None
Body-centered (<i>I</i>) Cubic	Body-Centered Cubic metal	$h+k+l$ even	$h+k+l$ odd
Face-centered (<i>F</i>) Cubic	Face-Centered Cubic metal	h,k,l all odd or all even	h,k,l mixed odd or even
Face-centered (<i>F</i>) Cubic	NaCl-rocksalt, ZnS-zincblende	h,k,l all odd or all even	h,k,l mixed odd or even
Face-centered (<i>F</i>) Cubic	C, Si, Ge - Diamond cubic	As FCC, but if all even then $h+k+l=4n$ (n is integer)	h,k,l mixed odd or even and if all even and $h+k+l \neq 4n$
Primitive (<i>P</i>) Hexagonal	Hexagonal closed packed metal	All other cases	$h+2k=3n$, l odd

Powder diffraction

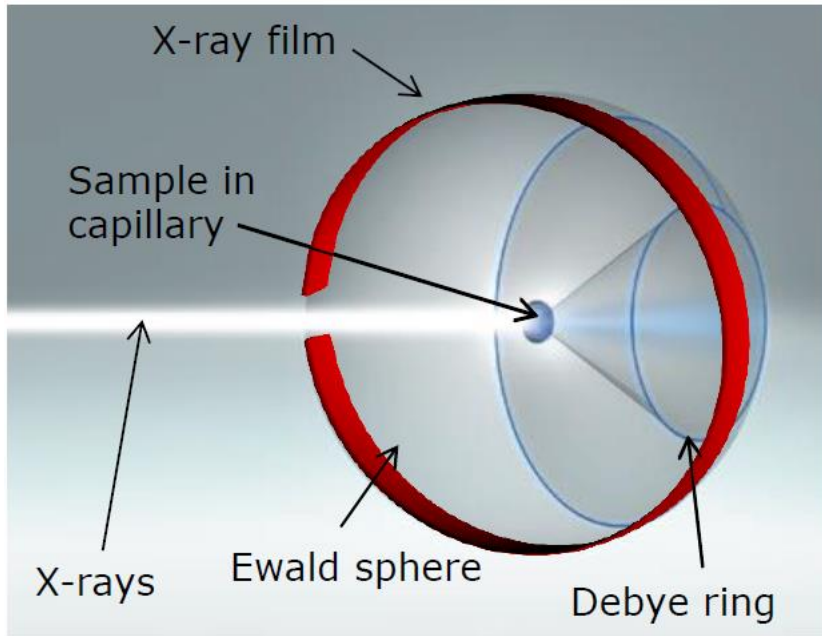
- Powders (i.e., polycrystalline aggregates) are billions of tiny crystallites in all possible orientations.
- When placed in an x-ray beam, all possible interatomic planes will be seen
- By systematically changing the experimental angle, we will produce all possible diffraction peaks from the powder



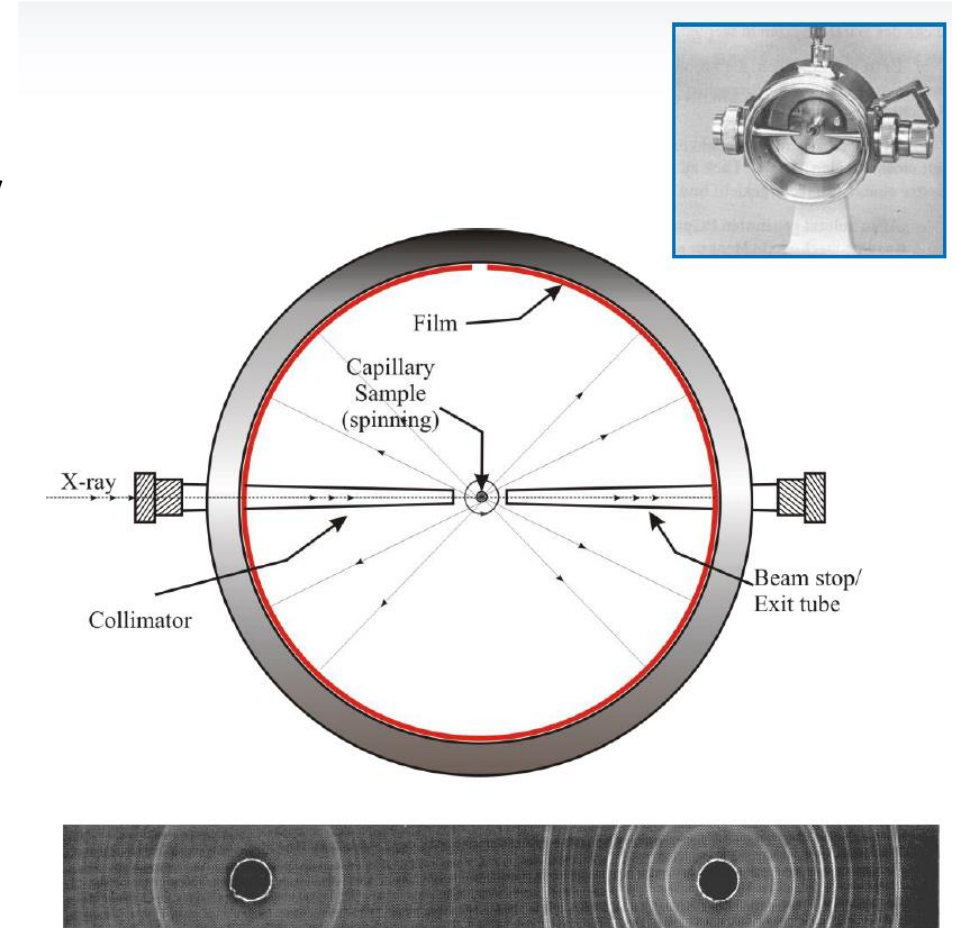
A “perfect” powder has:

- Infinite number of crystallites
- Infinite number of (random) orientations

Debye-Scherrer camera

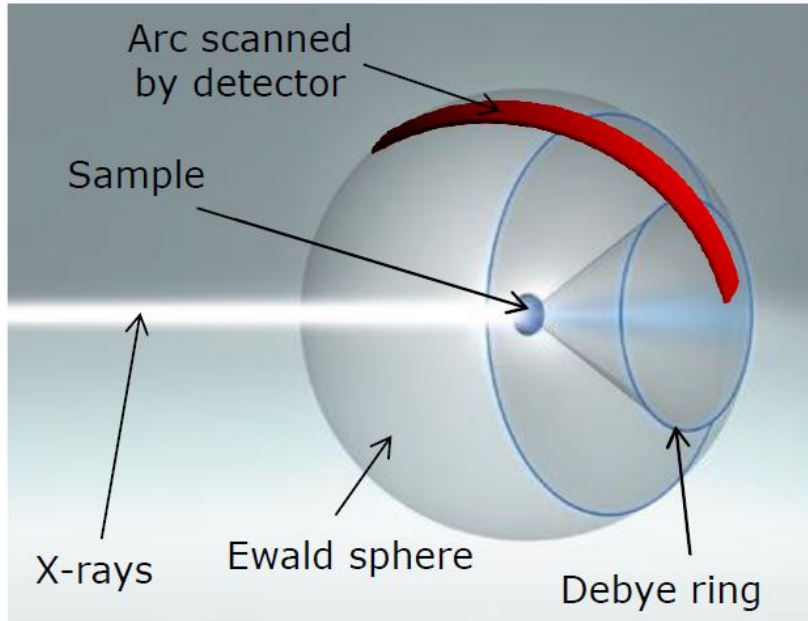


Transmission geometry

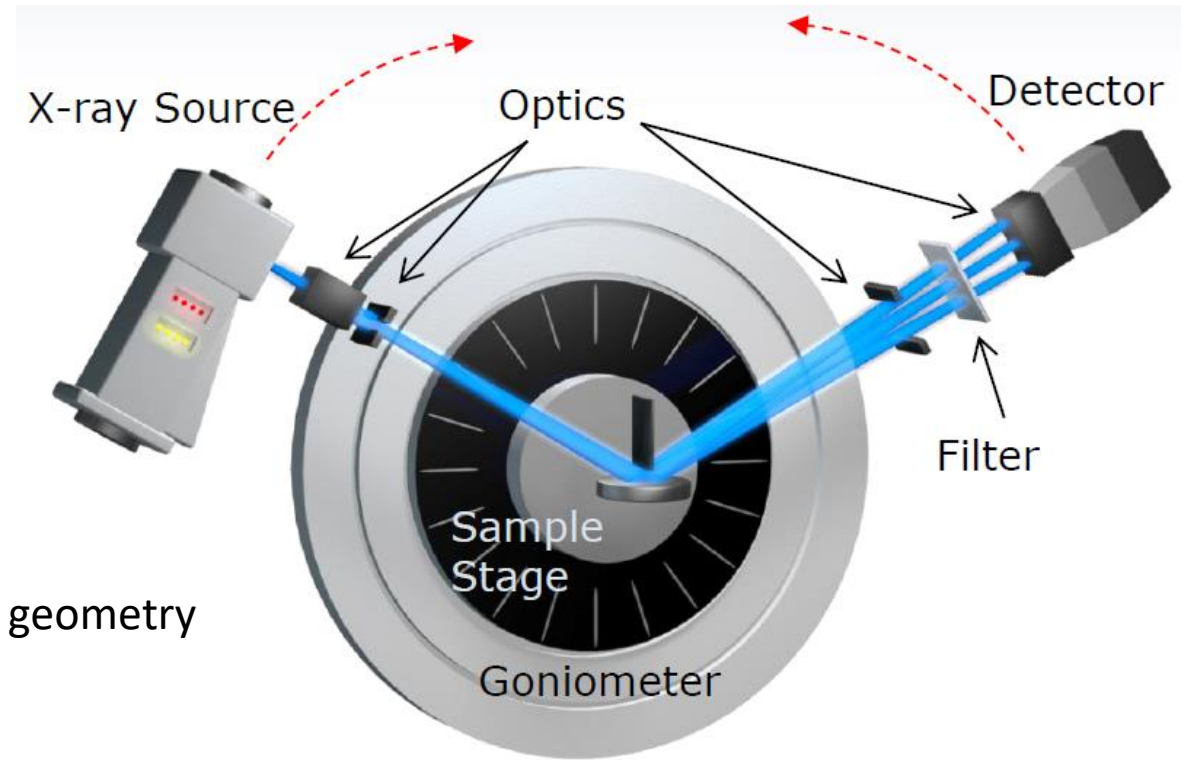


- Film is used as detector
- Collimators shield the film from radiation scattered by air
- 2θ positions read out manually

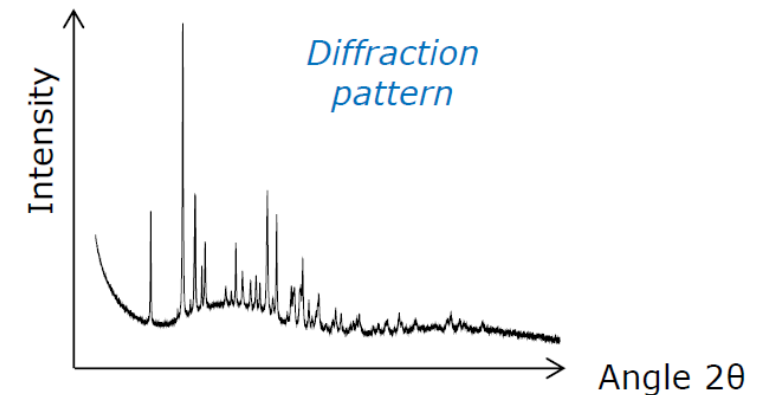
Bragg-Brentano geometry



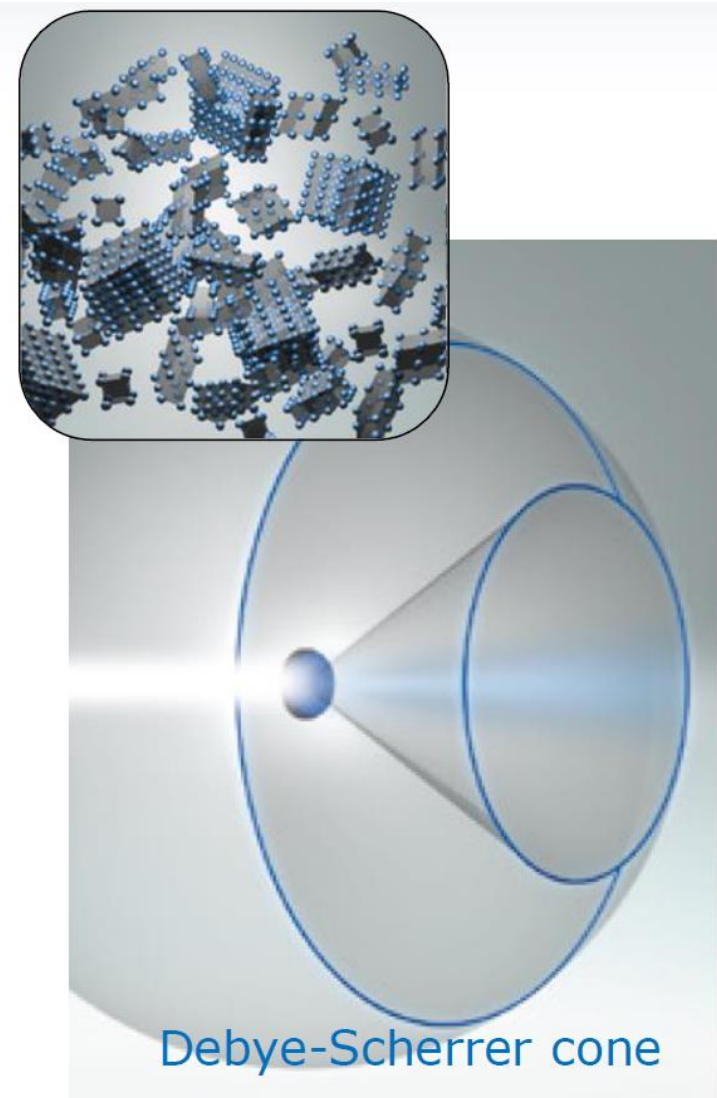
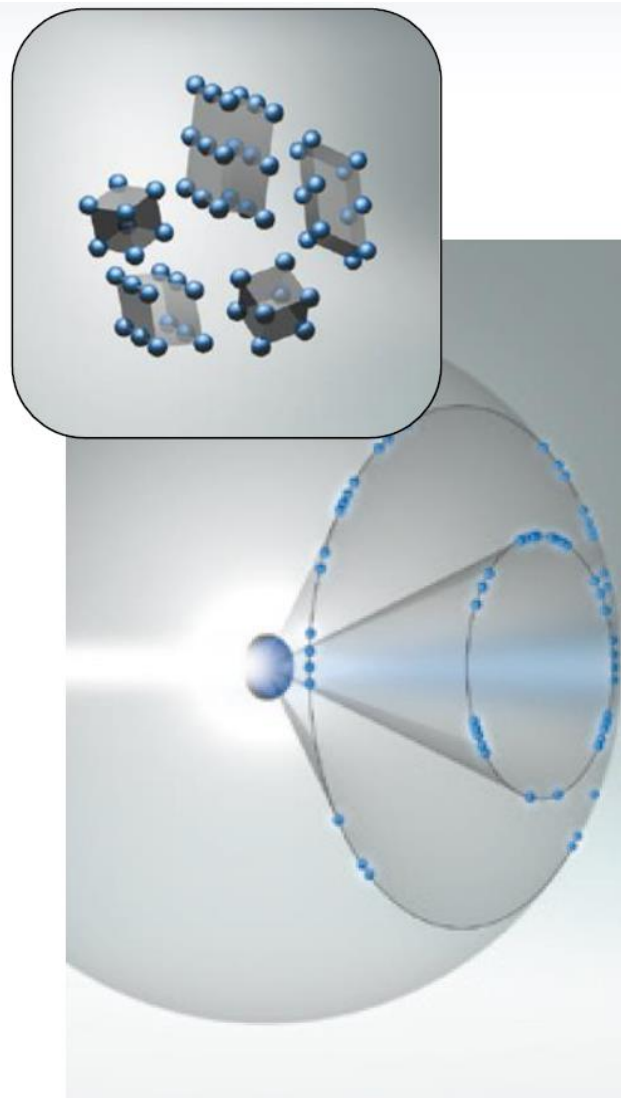
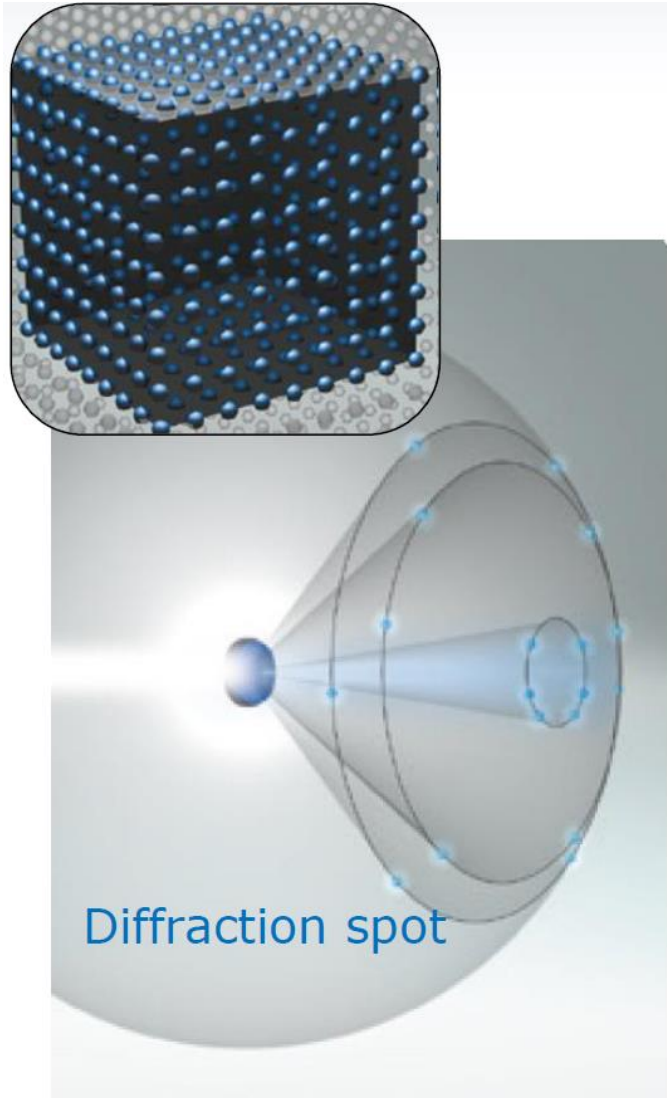
Reflection geometry



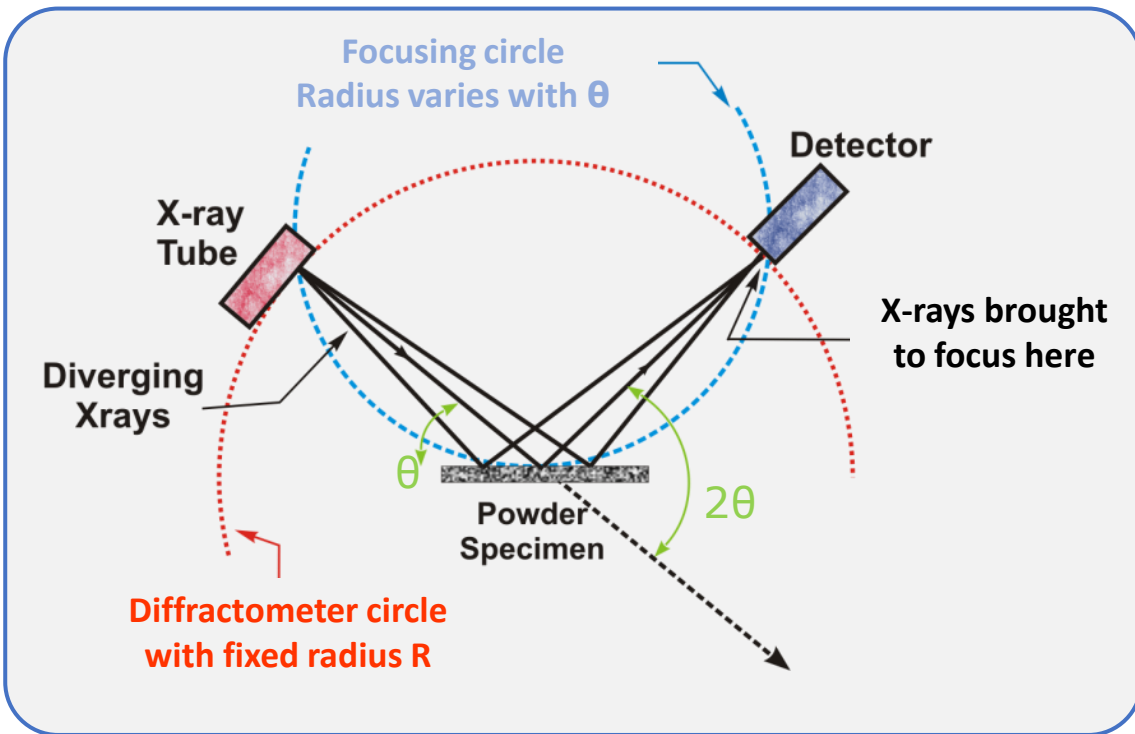
- X-ray source and detector move together
- The diffraction pattern is recorded with a detector and stored electronically
- Data is analyzed with dedicated software.



Diffraction from multiple crystals

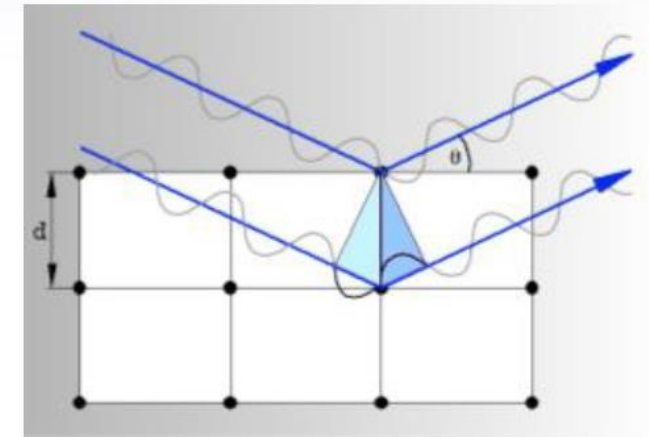


Bragg-Brentano geometry

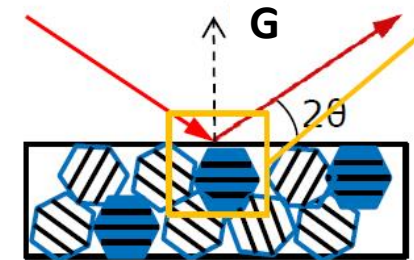


- The scattering vector ($\mathbf{G}$) is always normal to the sample surface.
- Only lattice planes normal to $\mathbf{G}$ (parallel to the sample surface) will be in the diffraction condition.

- The divergent beam from the X-ray source is reflected off the sample and then focused on the detector (**Reflection geometry**)
- Ideally, the sample surface would be curved at the radius of the focusing circle

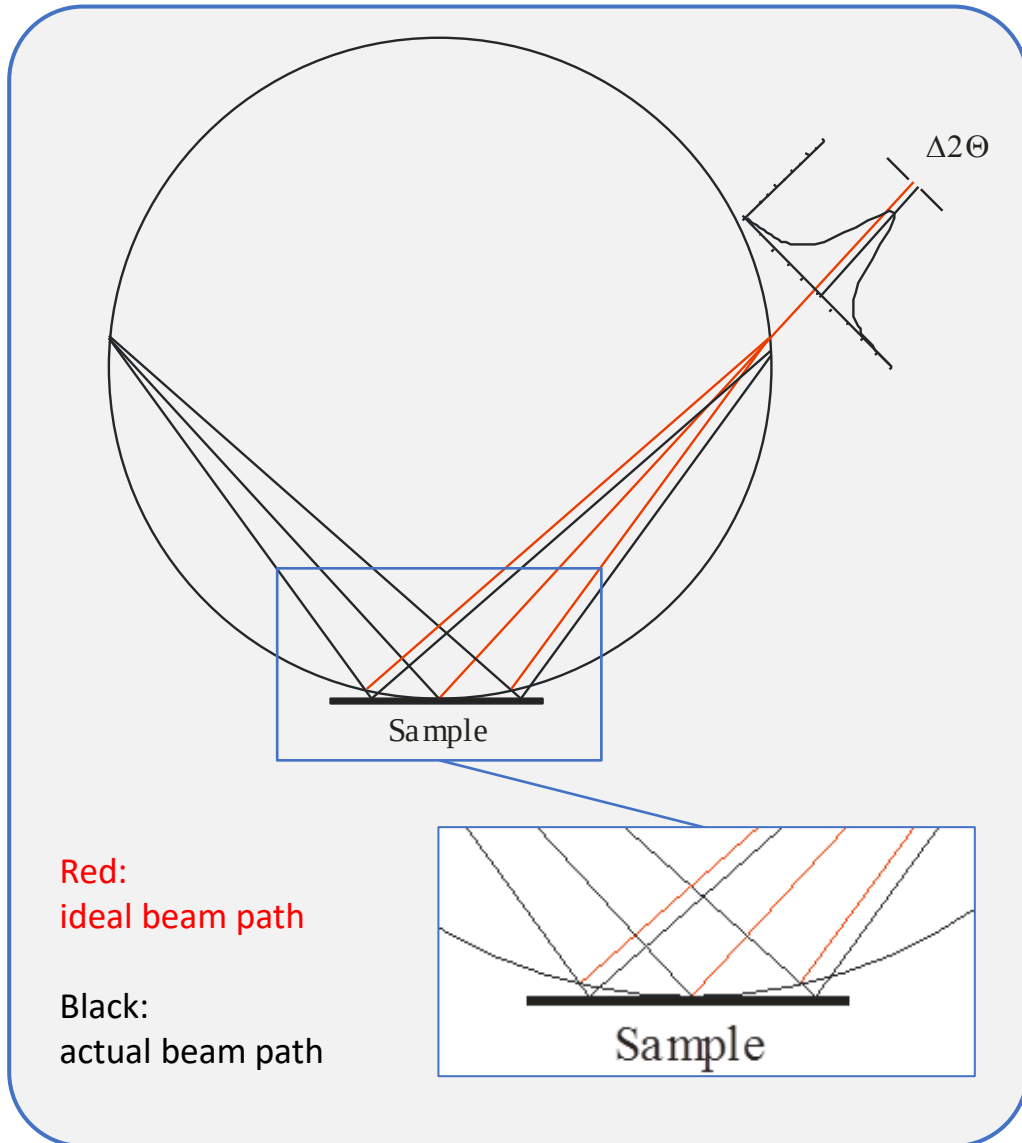


Constructive interference → Signal
Bragg's law: $n\lambda = 2d \sin(\theta)$



Only crystallites in the proper orientation will diffract

Flat specimen error

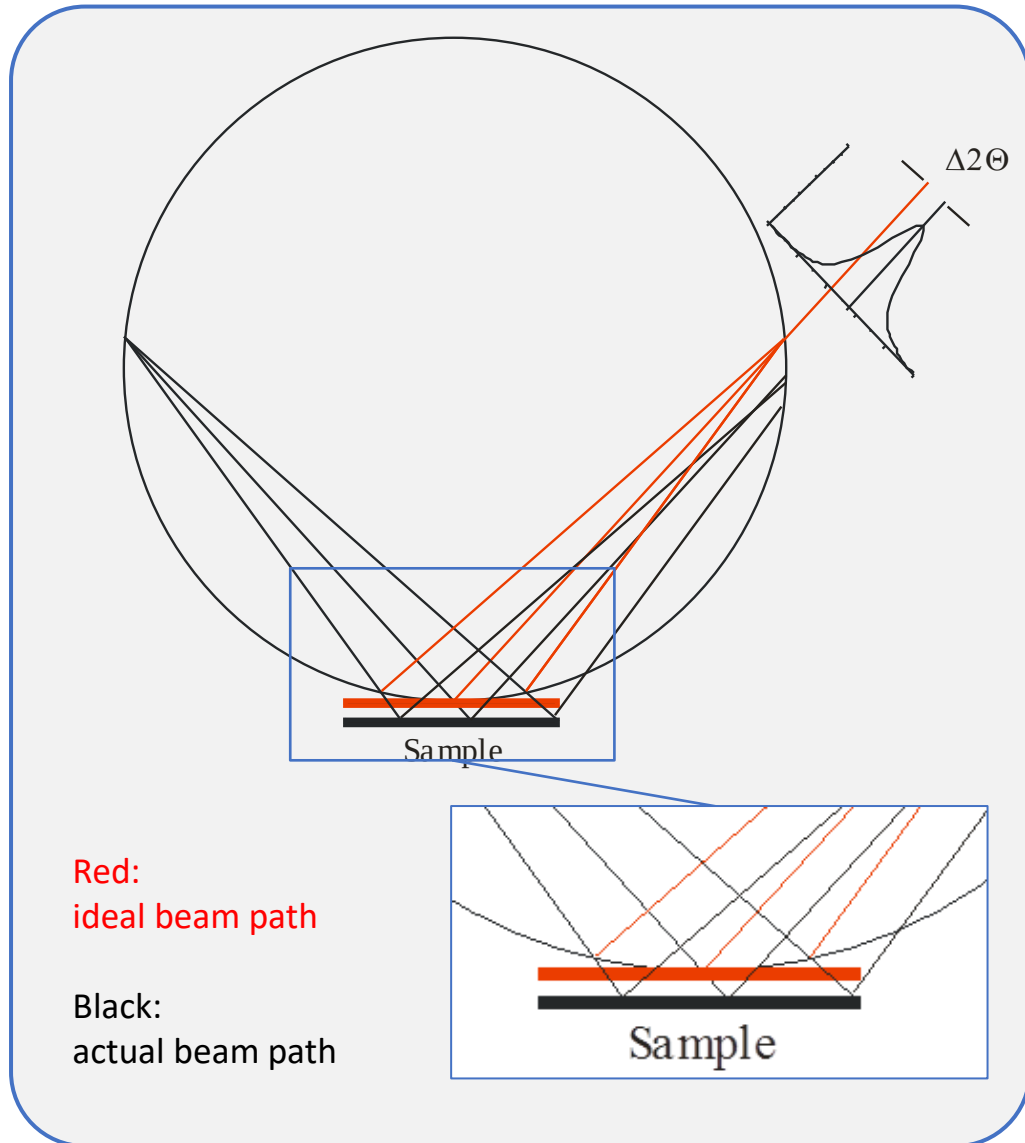


- In reality the the sample is not curve, but flat and tangent to the variable focusing circle.
- This leads to **peak shifts** and **asymmetric broadening**, because the X-rays are not exactly focused on the focusing circle.

$$\Delta 2\theta = \frac{-\alpha^2 \cot \theta}{343.8}$$

α : aperture of the divergence slit in degrees
 θ : in radians

Sample displacement error



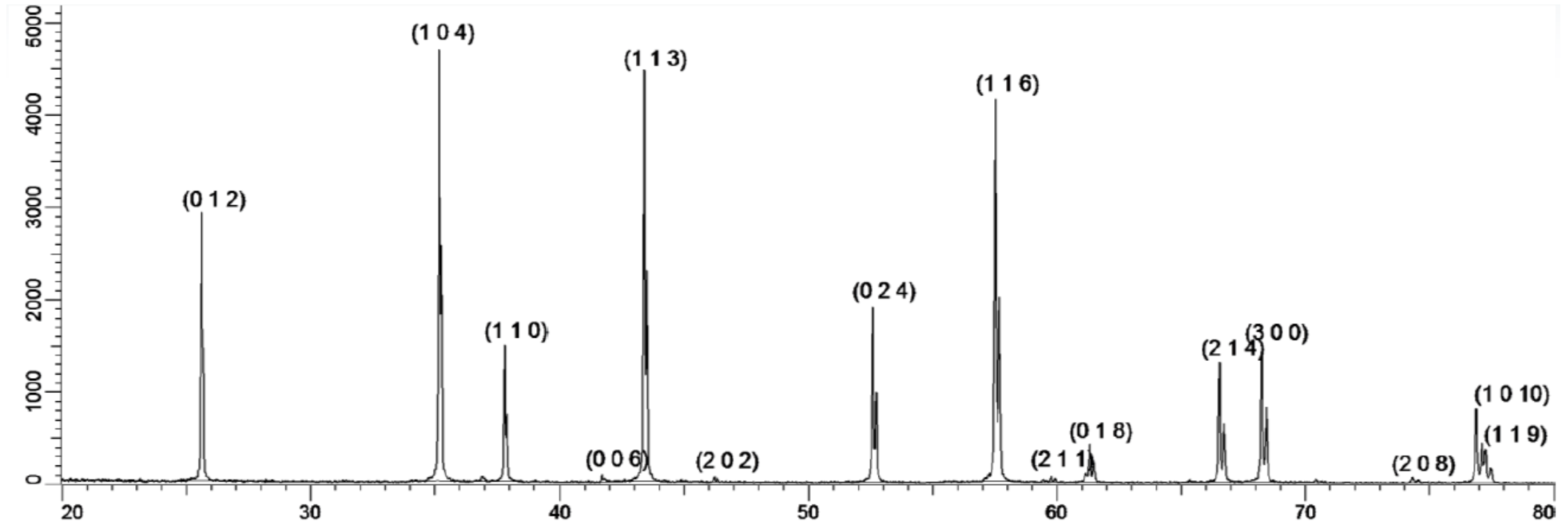
- The sample must be tangent to the focusing circle. Any deviations lead to **peak shifts** and **asymmetric broadening**
- Note: The sample displacement error is typically the largest error found in Bragg-Brentano geometry

$$\Delta 2\theta = \frac{-(114.59s \cdot \cos \theta)}{R} \quad \text{in degrees}$$

s: displacement

R: the goniometer radius

Powder diffraction



- Each peak of a pattern may be described by its Miller indexes
- The d-spacings of lattice planes depend on the size of the unit cell and determine the position of the peaks

in a cubic lattice

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

- The powder diffraction pattern of a compound is like a “fingerprint”: it can be used for identification

The value of d , the distance between adjacent planes in the set (hkl) , may be found from the following equations.

Cubic:
$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:
$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:
$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + hk + k^2}{a^2} \right) + \frac{l^2}{c^2}$$

Rhombohedral:

$$\frac{1}{d^2} = \frac{(h^2 + k^2 + l^2) \sin^2 \alpha + 2(hk + kl + hl)(\cos^2 \alpha - \cos \alpha)}{a^2(1 - 3 \cos^2 \alpha + 2 \cos^3 \alpha)}$$

Orthorhombic:
$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Monoclinic:
$$\frac{1}{d^2} = \frac{1}{\sin^2 \beta} \left(\frac{h^2}{a^2} + \frac{k^2 \sin^2 \beta}{b^2} + \frac{l^2}{c^2} - \frac{2hl \cos \beta}{ac} \right)$$

Triclinic:
$$\frac{1}{d^2} = \frac{1}{V^2} (S_{11}h^2 + S_{22}k^2 + S_{33}l^2 + 2S_{12}hk + 2S_{23}kl + 2S_{13}hl)$$

In the equation for triclinic crystals,

V = volume of unit cell (see below),

$$S_{11} = b^2c^2 \sin^2 \alpha,$$

$$S_{22} = a^2c^2 \sin^2 \beta,$$

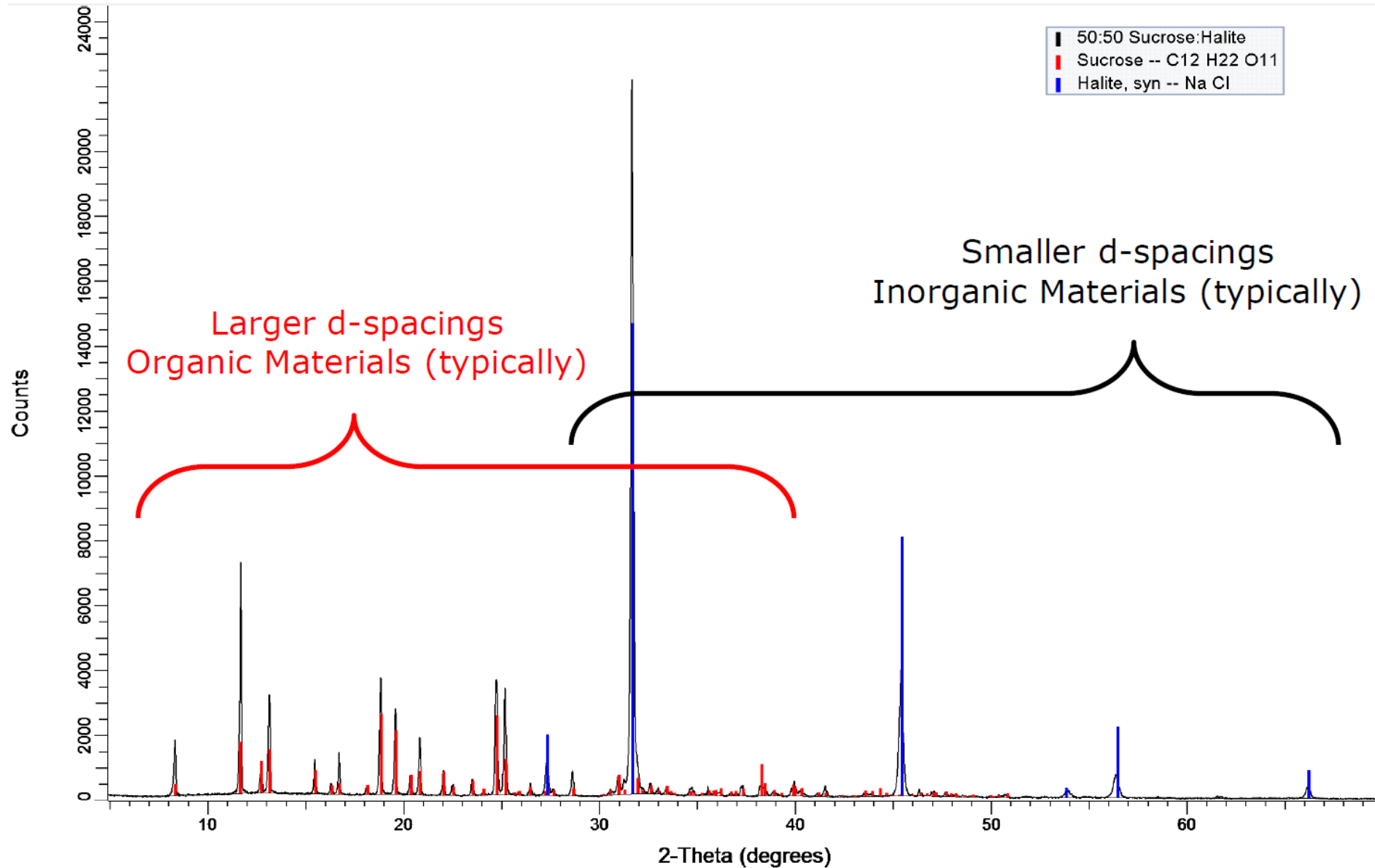
$$S_{33} = a^2b^2 \sin^2 \gamma,$$

$$S_{12} = abc^2(\cos \alpha \cos \beta - \cos \gamma),$$

$$S_{23} = a^2bc(\cos \beta \cos \gamma - \cos \alpha),$$

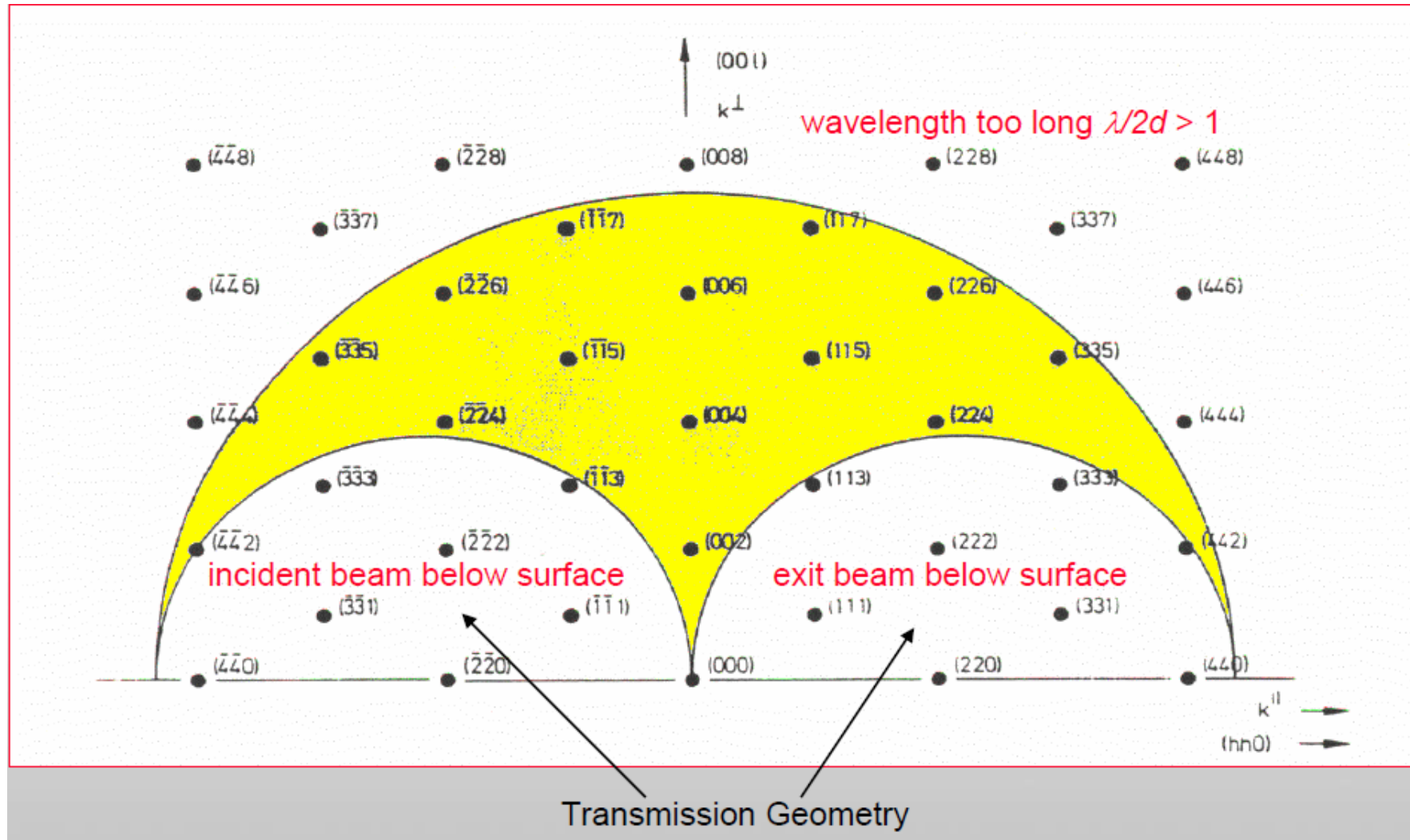
$$S_{13} = ab^2c(\cos \gamma \cos \alpha - \cos \beta).$$

Appropriate scanning range



$$2d \sin \theta = m\lambda$$

Ewald's sphere



Contributions to the diffraction pattern

Peak positions

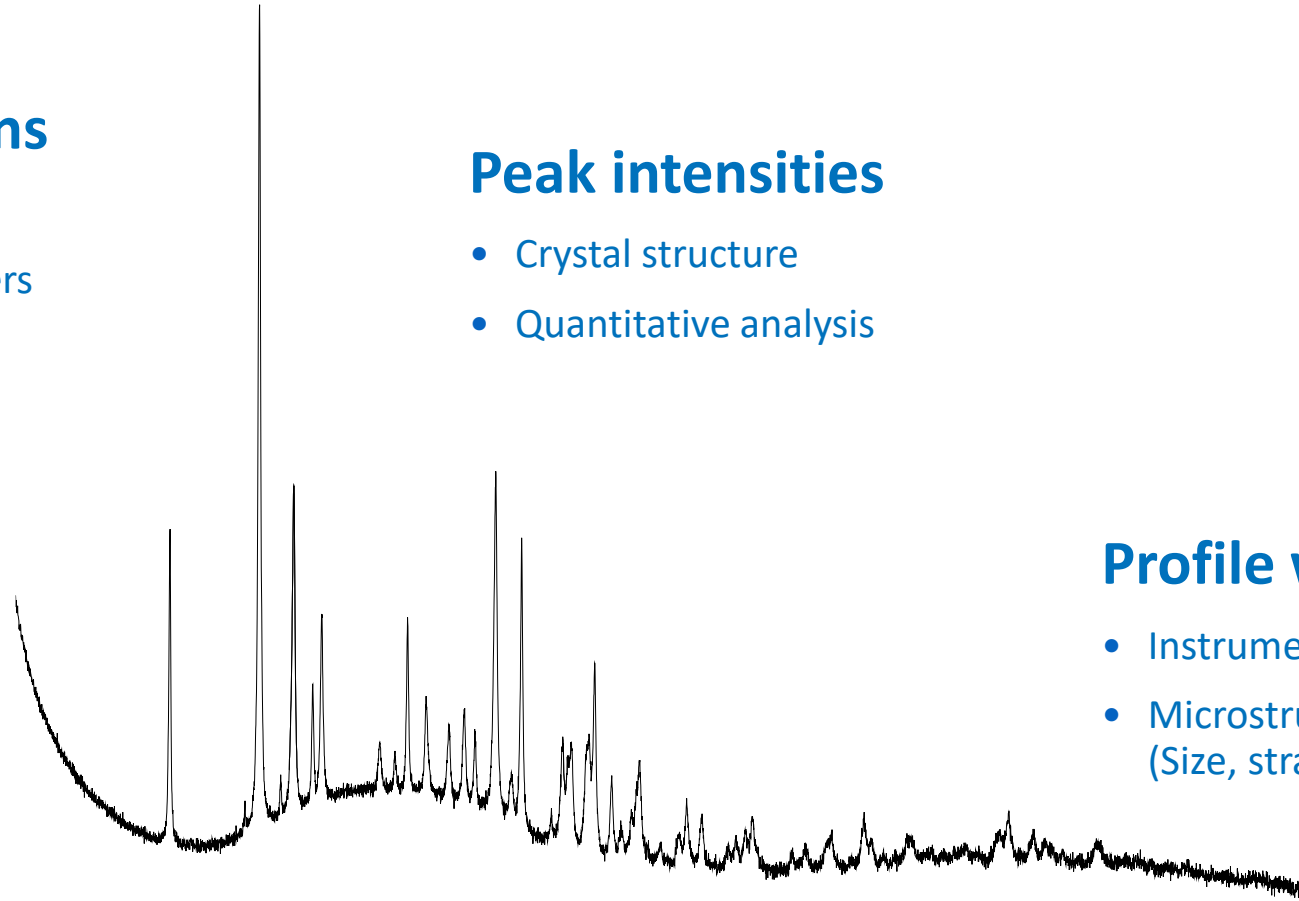
- Space group
- Lattice parameters

Peak intensities

- Crystal structure
- Quantitative analysis

Profile width and shape

- Instrument contributions
- Microstructure
(Size, strain, stacking faults, ...)

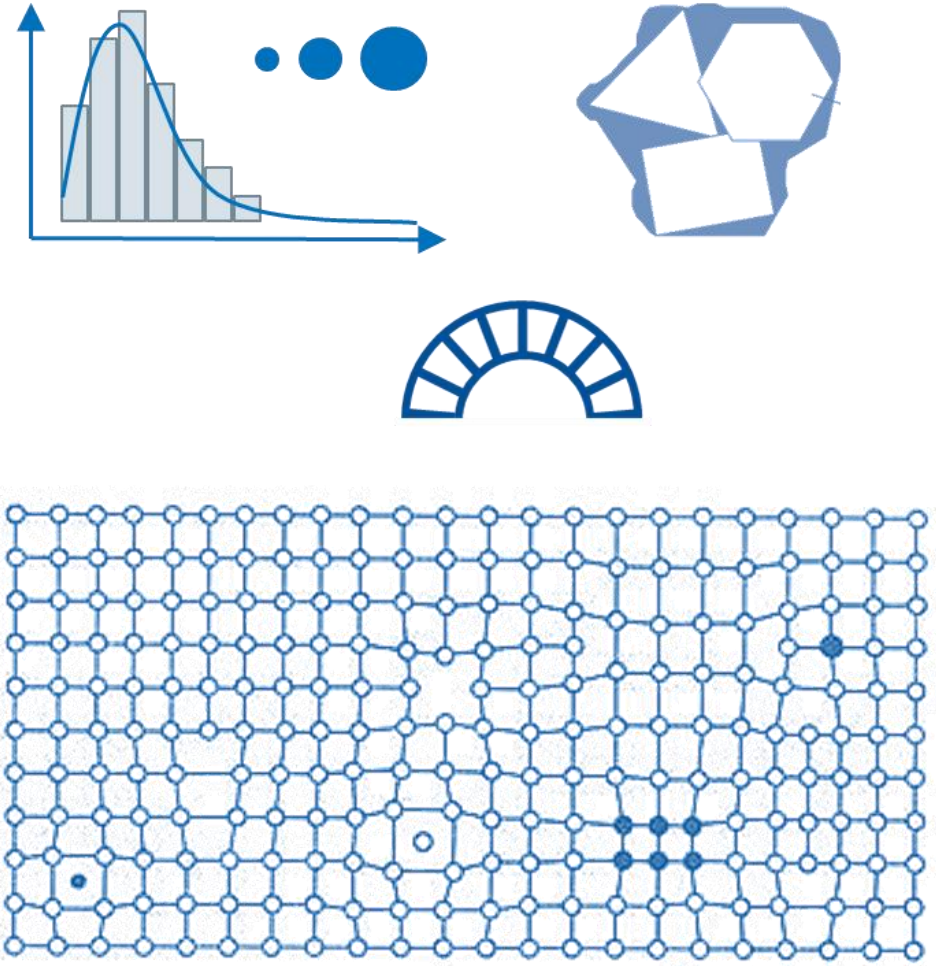


Background

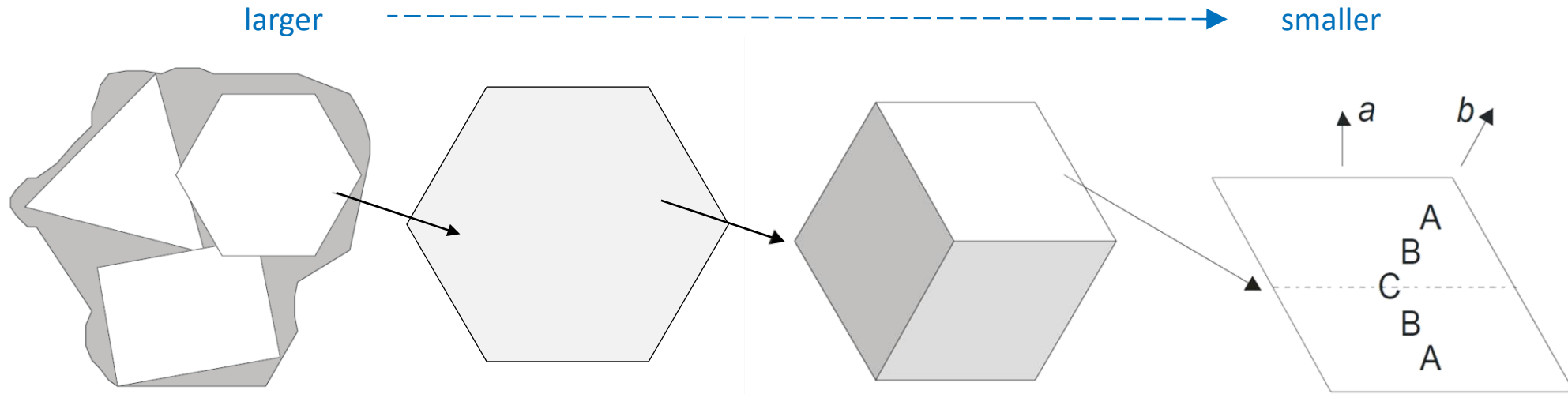
- Scattering from sample environment (air, sample holder, ...)
- Local order / disorder
- Amorphous phase amounts, "degree of crystallinity"

Origin of sample-related broadening

- Particle size (distribution)
- Particle shape (distribution)
- Strain
- Compositional variations
- Defects (vacancies, interstitial sites, dislocations, ...)
- Stacking faults



Size broadening: Terminology



Particle

Consists of several, separated crystals

Crystal

Infinite, 3D periodic lattice

Surface → 2D defect

Crystallite

Small crystals

Possibly held together through defective boundaries

Domains

Coherently diffracting volumes without 2D defects

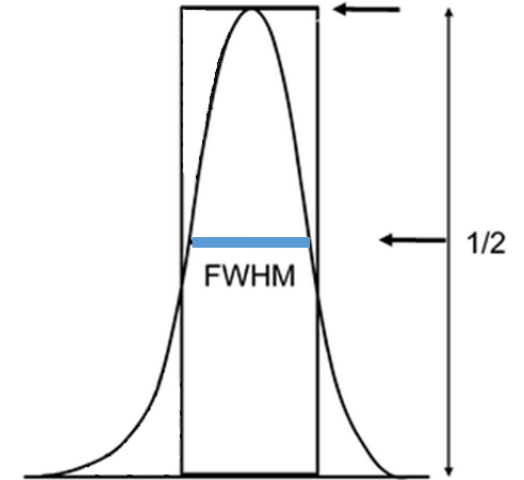
Size Broadening – Scherrer Equation

Scherrer (1918): Line profile width is inversely proportional to crystallite size.

$$"L_{Vol_FWHM}" = \frac{K\lambda}{\beta_{FWHM_s} \cdot \cos \theta}$$

"Scherrer size" (pointing to L_{Vol_FWHM})

peak broadening (pointing to β_{FWHM_s})



K: Scherrer “constant” (typically 0.87 – 1)

Depends on the shape of the crystal and the size distribution. Usually not known !

Commonly used *K* values:

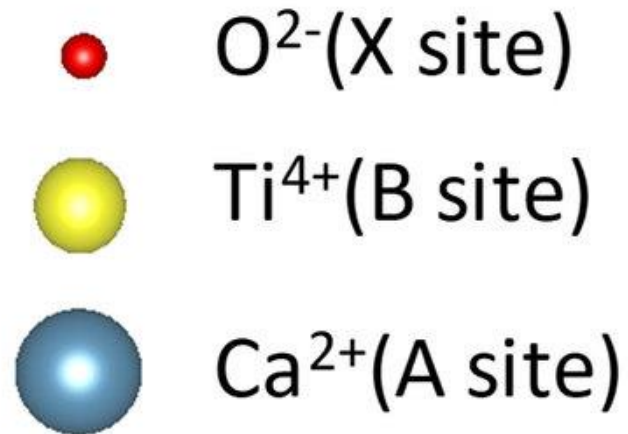
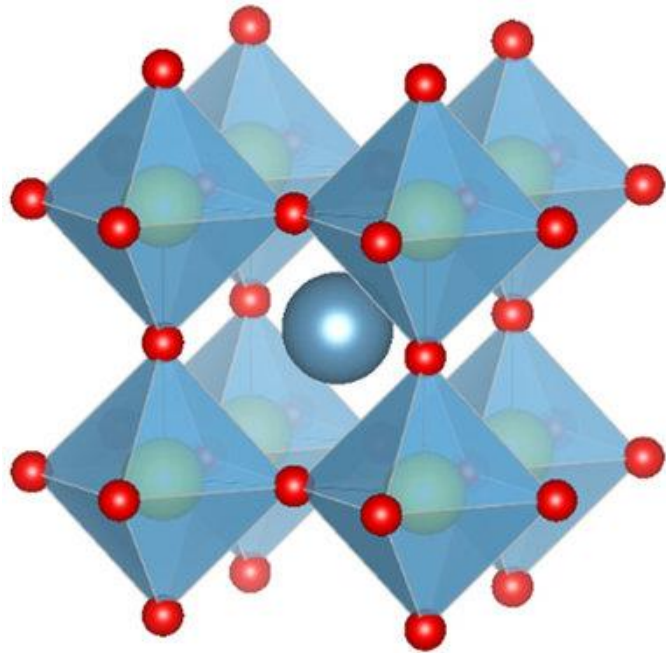
K = 0.89 for spherical crystallites (constant)

K ~ 0.94 for cubic crystallites (not constant; function of hkl)

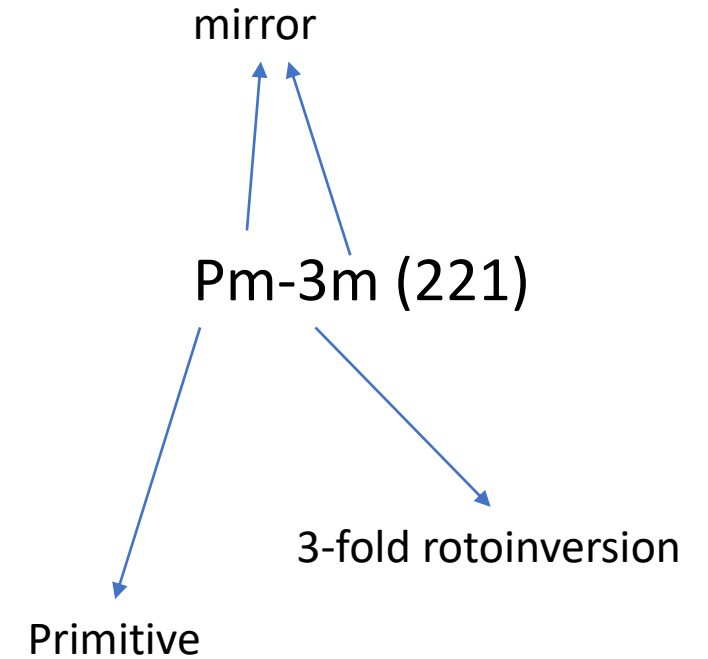
Only true if other microstructural effects are negligible!

Typically only gives a rough approximation

Perovskite structure



in general: ABX_3

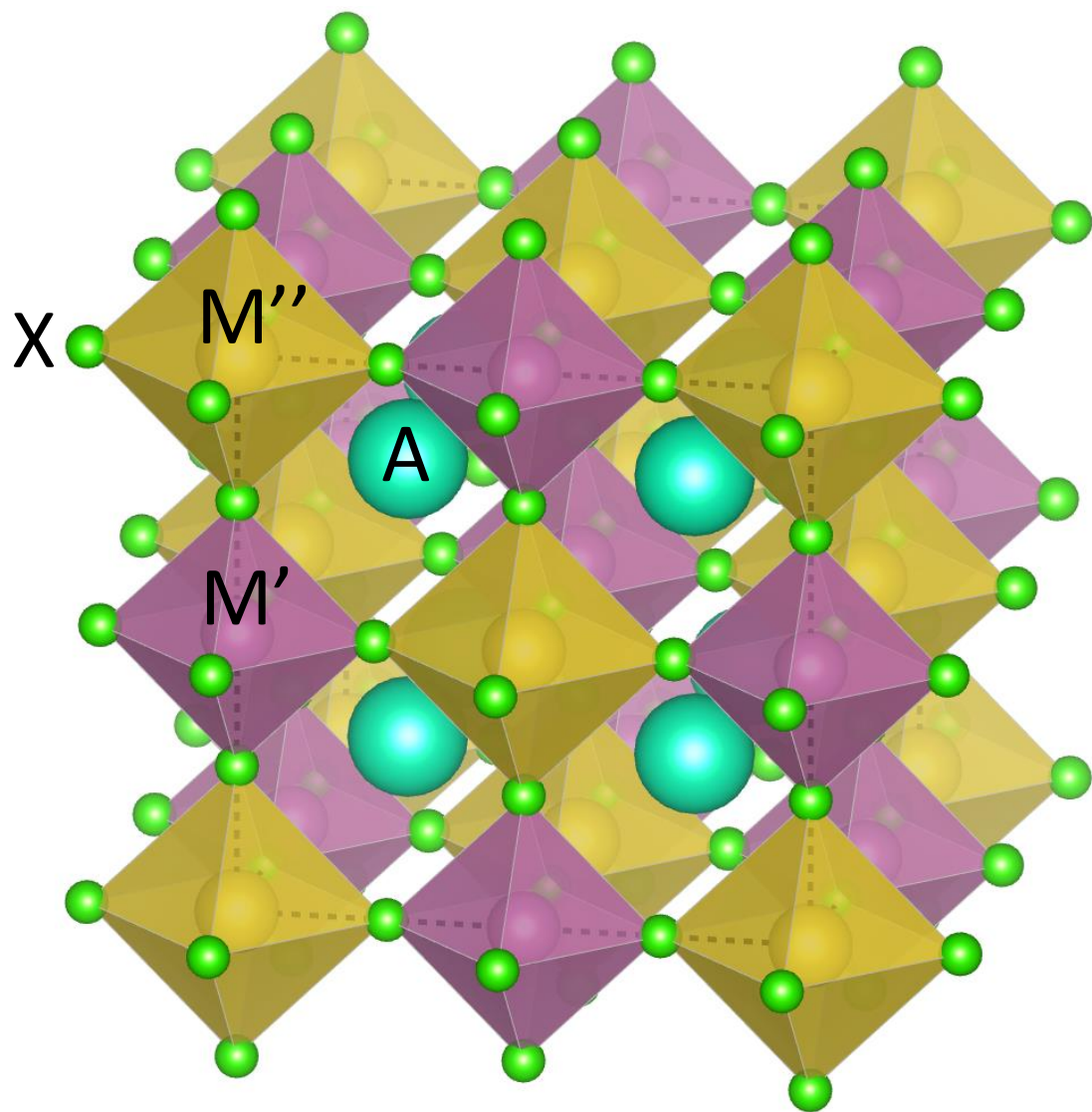


$$t = \frac{(r_A + r_X)}{\sqrt{2}(r_B + r_X)} \quad \text{tolerance factor}$$

$t = 1$ ideal cubic

$t < 1$ distorted

Double Perovskite $A_2M'M''X_6$



$\text{Cs}_2\text{AgInCl}_6$

Fm-3m 225

$\text{Cs}_2\text{NaInCl}_6$

Fm-3m 225

Experiment

Determinare i parametri reticolari e la composizione di tre polveri di perovskiti doppie del tipo $\text{Cs}_2\text{Na}_x\text{Ag}_{1-x}\text{InCl}_6$

1. Measure the XRD patterns of 3 powder samples
2. Rietveld analysis with Maud
3. Determination of x

